309 Final Project

Criminal Justice Data Analysis

Initial Disclaimer

The contents and conclusions of this study are limited in its scope, in part due to the rudimentary knowledge of exploratory data analysis of the researchers, as well as restricted time and resources to conduct analysis. Furthermore, the hypotheses and research direction are likely biased by the identities of the researchers, who are both educated women and are not representative of the diverse sample of respondents in this study. Lastly, this study only looks at a snapshot of data from one single year, 2023, and the incident variables picked out are only a small subset of the 1261 available variables - which is expanded upon in a larger ethical reflection below - but is important to recognize as a limitation before drawing any conclusions from the selected data.

```
load("~/CBSC 309/Criminal Justice Project/ICPSR_38962/DS0003/38962-0003-Data.rda")



load("~/CBSC 309/Criminal Justice Project/ICPSR_38962/DS0004/38962-0004-Data.rda")


library(tidyverse)
library(psych)
library(yarrr)
```

Dataset Description

The National Crime Victimization Survey (NCVS) has collected annual data on personal and property crime since 1973. This survey is conducted by the Bureau of Justice Statistics, and each respondent is asked a series of screen questions designed to determine whether they were victimized during the six month period preceding the first day of the month of the interview. This version of the NCVS, referred to as the collection year, contains records from interviews conducted in the 12 months of the given year, 2023.

The NCVS file extracted from the National Archive of Criminal Justice Data (NACJD) website contains the following four datasets. First, the "Address Record" dataset which contains household addresses, year, sample number and person ID - 255,901 observations. Second, the "Household Record" dataset contains the same variables as the address record, in addition to information about land ownership, usage, residents, physical attributes etc. and has 254,071 observations. Third, the "Person Record" dataset contains demographic information of the participant, including variables of interest like race, sex, gender and educational achievement of the participant. Lastly, the "Incident Record" dataset contains information about the nature of the incident, or criminal activity, as well as pre and post data such as perceived cause, injuries, impact of incident on work, health, etc. This dataset contains 9324 observations.

This study selected the "Person Record" and "Incident Record" datasets to analyze. The survey categorizes crimes as "personal" or "property". Personal crimes include rape and sexual attack, robbery, aggravated and simple assault and purse-snatching/pocket-picking, white property crimes include burglary, theft, motor vehicle theft and vandalism. We will focus on personal crime to answer the question - *how does gender affect the likelihood of being a victim of violent personal crime?* It is important to note that this data codes gender as a binary value (Male and Female) rather than as it exists on a spectrum in reality.

```
person_data <- da38962.0003 %>%
  select(IDPER,
         V3018,
         V3085,
         V3086,
         V3020,
         V3023A,
         V3024)
```

```
incident_data <- da38962.0004 %>%
  select(IDPER,
         V4012,
         V4016,
         V4247,
         V4097,
         V4100,
         V4102,
         V4103,
         V4104,
         V4109,
         V4105,
         V4106,
         V4094,
         V4096,
         V4069,
         V4084,
         V4086,
         V4087,
         V4088,
         V4079,
         V4090,
         V4080,
         V4089,
         V4078,
         V4095,
         V4081,
         V4082)

dataset <-  full_join(person_data, incident_data, join_by(IDPER), relationship = "many-to-many")


dataset_rename <- dataset %>%
  rename("Sex Allocated" = V3018,
         "Gender at Birth" = V3085,
         "Gender Current" = V3086,
         "Educational Attainment" = V3020,
         "Race Recoded" = V3023A,
         "Incident Number" = V4012,
         "How Many Incidents" = V4016,
         "Single Off Crime" = V4247,
```

```
        "Shot" = V4097,
        "Stabbed with Knife" = V4100,
        "Hit with Handheld Object" = V4102,
        "Hit with Thrown Object" = V4103,
        "Attack W Other Weapon" = V4104,
        "Threat and Attack" = V4109,
        "Battery: Hit or Slap" = V4105,
        "Battery: Grab or Trip" = V4106,
        "Raped" = V4094,
        "Sexual Assault" = V4096,
        "Sexual Contact w Force" = V4069,
        "Hispanic" = V3024) %>%
  mutate("How Many Incidents" =
          ifelse(`How Many Incidents` > 6, 6, `How Many Incidents`)) %>%
  mutate(`Race Recoded` = substr(`Race Recoded`, 2,3))


dataset_rename <- dataset_rename %>%
  mutate(`Race Recoded` = ifelse(`Race Recoded` == "01", 'White', `Race Recoded`))%>%
  mutate(`Race Recoded` = ifelse(`Race Recoded` == "02", 'Black', `Race Recoded`)) %>%
  mutate(`Race Recoded` = ifelse(`Race Recoded` == "05" , 'Native', `Race Recoded`)) %>%
  mutate(`Race Recoded` = ifelse(`Hispanic` == "(1) Yes", "Hispanic", `Race Recoded`)) %>%
  mutate(`Race Recoded` = ifelse(`Race Recoded` == "03", "Native", `Race Recoded`)) %>%
  mutate(`Race Recoded` = ifelse(`Race Recoded` == "04" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "06" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "07" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "08" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "09" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "10" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "11" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "12" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "13" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "14" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "15" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "16" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "17" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "18" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "19" , 'Mixed', `Race Recoded`)) %>%
   mutate(`Race Recoded` = ifelse(`Race Recoded` == "20" , 'Mixed', `Race Recoded`))
```

Incidents Variable Model:

On reviewing the codebook we decided to join together 22 categorical variables that were measured on a yes/no/NA scale to form larger continuous variables that could be analyzed. The continuous variables were named in accordance with the language used in the field of criminal justice, and are made up of the following categorical variables.

1. Attempted Aggravated Battery

i. Attempt with weapon present
ii. Attempt with knife
iii. Attempt with other weapon

iv. Attempt: Object Thrown
v. Attempt: Threat to Kill

2. Attempted Battery/Other

i. Attempt: Hit, Grab, Jump
ii. Attempt: Other
iii. Attempt: Followed or Surrounded

3. Attempted Sexual Crime

i. Attempt: Threat of rape
ii. Attempt: Threat of Sexual Assault
iii. Attempt: Sex Contact with Force

```
## Attempted Aggravated Battery

dataset_1 <- dataset %>%
  mutate(V4079 = na_if(V4079, "(8) Residue"),
         V4079 = ifelse(V4079 == '(1) Yes', 1, 0)) %>%
  mutate(V4084 = na_if(V4084, "(8) Residue"),
         V4084 = ifelse(V4084 == '(1) Yes', 1, 0)) %>%
   mutate(V4086 = na_if(V4086, "(8) Residue"),
         V4086 = ifelse(V4086 == "(1) Yes", 1, 0)) %>%
  mutate(V4087 = na_if(V4087, "(8) Residue"),
         V4087 = ifelse(V4087 == "(1) Yes", 1, 0)) %>%
   mutate(V4088 = na_if(V4088, "(8) Residue"),
         V4088 = ifelse(V4088 == "(1) Yes", 1, 0))


dataset_1 <- dataset_1 %>%
  mutate(sum_1 = V4079 + V4084 + V4086 + V4087 + V4088) %>%
  mutate(att_agg_batt_avg = sum_1/5)

dataset_1A <- dataset_1 %>%
  select(-V4079,
         -V4084,
         -V4086,
         -V4087,
         -V4088,
         -sum_1)

## Attempted Battery

dataset_2 <- dataset_1A %>%
  mutate(V4090 = na_if(V4090, "(8) Residue"),
         V4090 = ifelse(V4090 == '(1) Yes', 1, 0)) %>%
  mutate(V4080 = na_if(V4080, "(8) Residue"),
         V4080 = ifelse(V4080 == '(1) Yes', 1, 0)) %>%
  mutate(V4089 = na_if(V4089, "(8) Residue"),
         V4089 = ifelse(V4089 == '(1) Yes', 1, 0))
```

```r
dataset_2 <- dataset_2 %>%
  mutate(sum_1 = V4089 + V4090 + V4080) %>%
  mutate(att_batt_avg = sum_1/3)

dataset_2A <- dataset_2 %>%
  select(-V4089,
         -V4080,
         -V4090,
         -sum_1)

## Attempted Sexual Crime

dataset_3 <- dataset_2 %>%
  mutate(V4078 = na_if(V4078, "(8) Residue"),
         V4078 = ifelse(V4078 == '(1) Yes', 1, 0)) %>%
  mutate(V4081 = na_if(V4081, "(8) Residue"),
         V4081 = ifelse(V4081 == '(1) Yes', 1, 0)) %>%
  mutate(V4082 = na_if(V4082, "(8) Residue"),
         V4082 = ifelse(V4082 == '(1) Yes', 1, 0))

dataset_3 <- dataset_3 %>%
  mutate(sum_1 = V4078 + V4081 + V4082) %>%
  mutate(att_SA_avg = sum_1/3)

dataset_3A <- dataset_3 %>%
  select(-V4078,
         -V4081,
         -V4082,
         -sum_1)
```

4. Aggravated Battery

i. Shot
ii. Stabbed/Cut with Knife
iii. Hit by Object held in hand
iv. Hit by Thrown Object
v. Attack with Other Weapon
vi. Threat and then Attack

5. Battery

i. Hit, slapped, knocked down
ii. Grabbed, held, tripped etc.

6. Sexual Crime

i. Rape
ii. Sexual Assault
iii. Tried to rape

```r
## Aggravated Battery

dataset_4 <- dataset_3A %>%
  mutate(V4097 = na_if(V4097, "(8) Residue"),
         V4097 = ifelse(V4097 == '(1) Yes', 1, 0)) %>%
  mutate(V4100 = na_if(V4100, "(8) Residue"),
         V4100 = ifelse(V4100 == '(1) Yes', 1, 0)) %>%
  mutate(V4102 = na_if(V4102, "(8) Residue"),
         V4102 = ifelse(V4102 == '(1) Yes', 1, 0)) %>%
  mutate(V4103 = na_if(V4103, "(8) Residue"),
         V4103 = ifelse(V4103 == '(1) Yes', 1, 0)) %>%
  mutate(V4104 = na_if(V4104, "(8) Residue"),
         V4104 = ifelse(V4104 == '(1) Yes', 1, 0)) %>%
  mutate(V4109 = na_if(V4109, "(8) Residue"),
         V4109 = ifelse(V4109 == '(1) Yes', 1, 0))

dataset_4 <- dataset_4 %>%
  mutate(sum_1 = V4097 + V4100 + V4102 + V4103 + V4104 + V4109) %>%
  mutate(agg_bat_avg = sum_1/6)

dataset_4A <- dataset_4 %>%
  select(-V4097,
         -V4100,
         -V4102,
         -V4103,
         -V4104,
         -V4109,
         -sum_1)

## Battery
dataset_5 <- dataset_4A %>%
  mutate(V4105 = na_if(V4105, "(8) Residue"),
         V4105 = ifelse(V4105 == '(1) Yes', 1, 0)) %>%
  mutate(V4106 = na_if(V4106, "(8) Residue"),
         V4106 = ifelse(V4106 == '(1) Yes', 1, 0))

dataset_5 <- dataset_5 %>%
  mutate(sum_1 = V4105 + V4106) %>%
  mutate(bat_avg = sum_1/2)

dataset_5A <- dataset_5 %>%
  select(-V4105,
         -V4106,
         -sum_1)

## Sexual Crimes

dataset_6 <- dataset_5A %>%
  mutate(V4094 = na_if(V4094, "(8) Residue"),
         V4094 = ifelse(V4094 == '(1) Yes', 1, 0)) %>%
  mutate(V4095 = na_if(V4095, "(8) Residue"),
         V4095 = ifelse(V4095 == '(1) Yes', 1, 0)) %>%
  mutate(V4096 = na_if(V4096, "(8) Residue"),
```

```
        V4096 = ifelse(V4096 == '(1) Yes', 1, 0))


dataset_6 <- dataset_6 %>%
  mutate(sum_1 = V4094 + V4096 + V4095 ) %>%
  mutate(SA_avg = sum_1/3)

dataset_6A <- dataset_6 %>%
  select(-V4094,
         -V4096,
         -V4095,
         -V4069,
         -sum_1)
```

Data visualisation

Number of Incidents ~ Sex

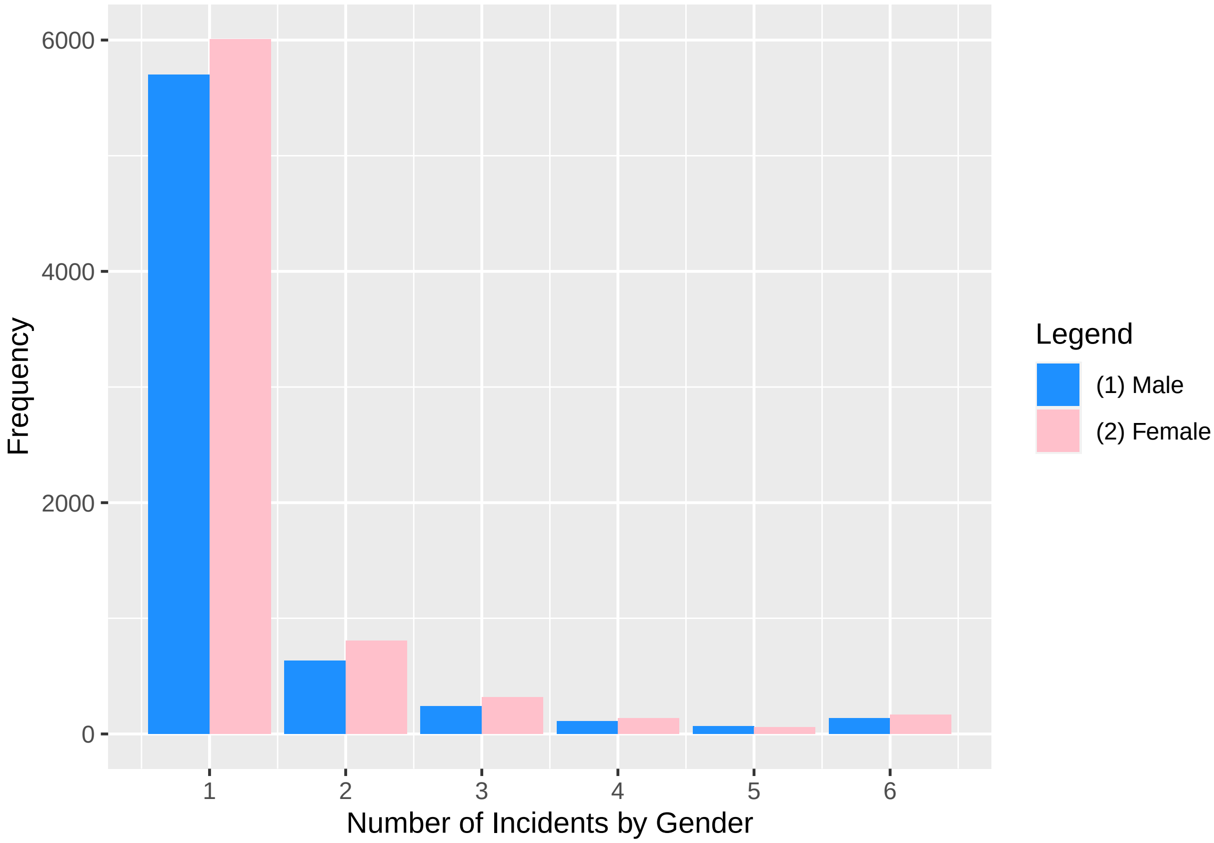
The Incidents plots show the frequency with which a respondent reported being victim of a violent personal crime a given number of times. In the data, the maximum number of instances of victimization a victim could report was 10+, however, due to the scarcity of responses in the higher numbers of incident responses, we decided to aggregate victims who had been victimized 6 or more times in the 6 (and greater) category.

The first bar graph shows the raw frequency of each response separated by binary gender. Raw frequency scales from 0 to greater than 6000, given that there were almost 280,000 observations our dataset. The graph is skewed towards the bottom, meaning that there are much more people who experience less than 3 victimizations than those who experience more than 3 victimizations, regardless of gender. Women are slightly more likely to have experienced 1, 2, or 3 victimizations. In victims above 3 victimizations, it seems there is no difference in responses between males and females.

The second bar graph shows the raw frequency of each response separated by race. This graph is also skewed towards the bottom, meaning that there are much more people who experience less than 3 victimizations than those who experience more than 3 victimizations, regardless of gender. Across the scale, white people reported a higher frequency of victimizations, followed by hispanic and black people. This does not indicate that white people have a higher likelihood of being victims, more that this sample has an overrepresentation of white respondents as compared to black, hispanic, native american or mixed race respondents. This is further evidenced in the subsequent pirateplot that shows a summary view of incidents, where white people report about the same number of incidents as other races, and native american respondents report a slightly higher number of incidents, although the relatively wide error margins around the mean imply that this is not reliable.

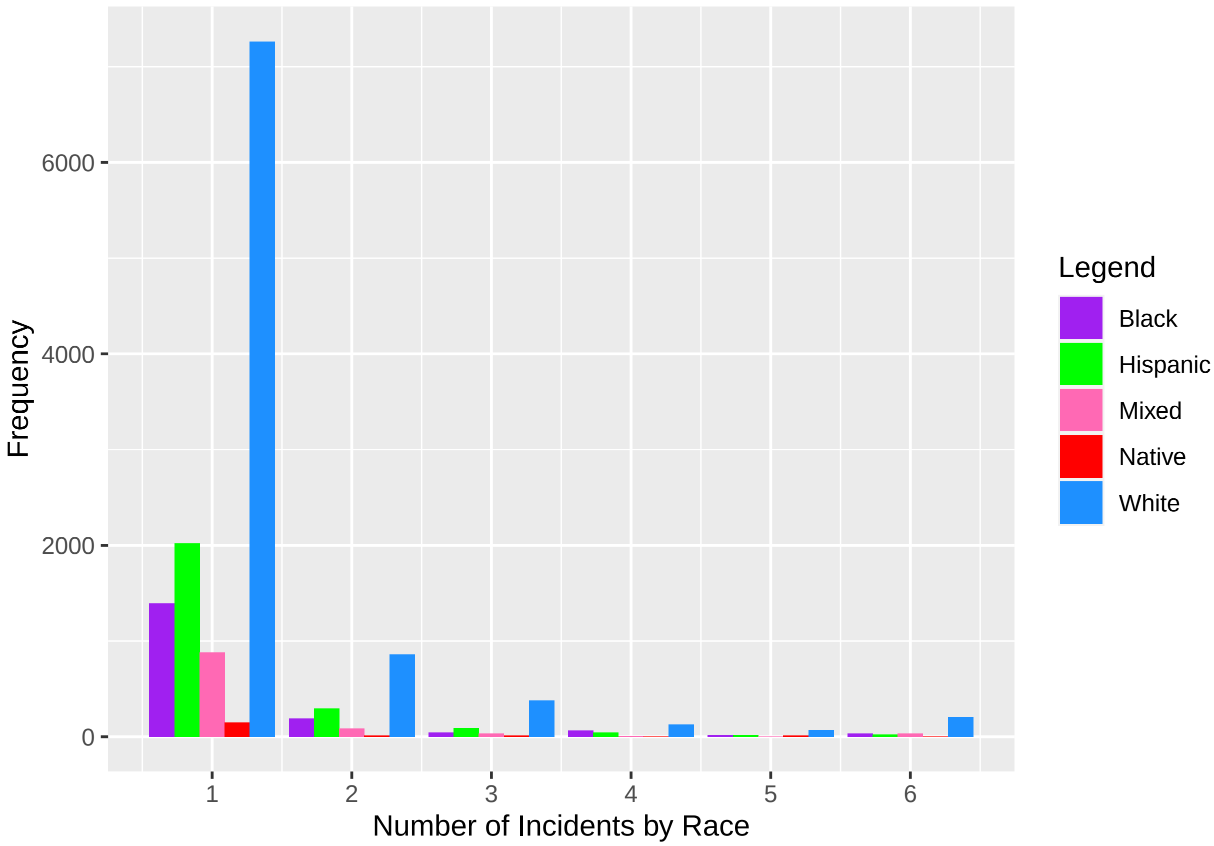
```
incident_plot <-ggplot(dataset_rename,
                       aes(x = `How Many Incidents`,
                           fill = `Sex Allocated`))+
  geom_bar(stat = "count",
           position = position_dodge()) +
  scale_fill_manual('Legend',
                    values = c('(1) Male' = 'dodgerblue', '(2) Female' = 'pink')) +
  scale_x_continuous(breaks = round(seq(min(0), max(6), by = 1),1)) +
  xlab("Number of Incidents by Gender") +
  ylab("Frequency")

incident_plot
```

```
## Warning: Removed 264395 rows containing non-finite values (`stat_count()`).
```

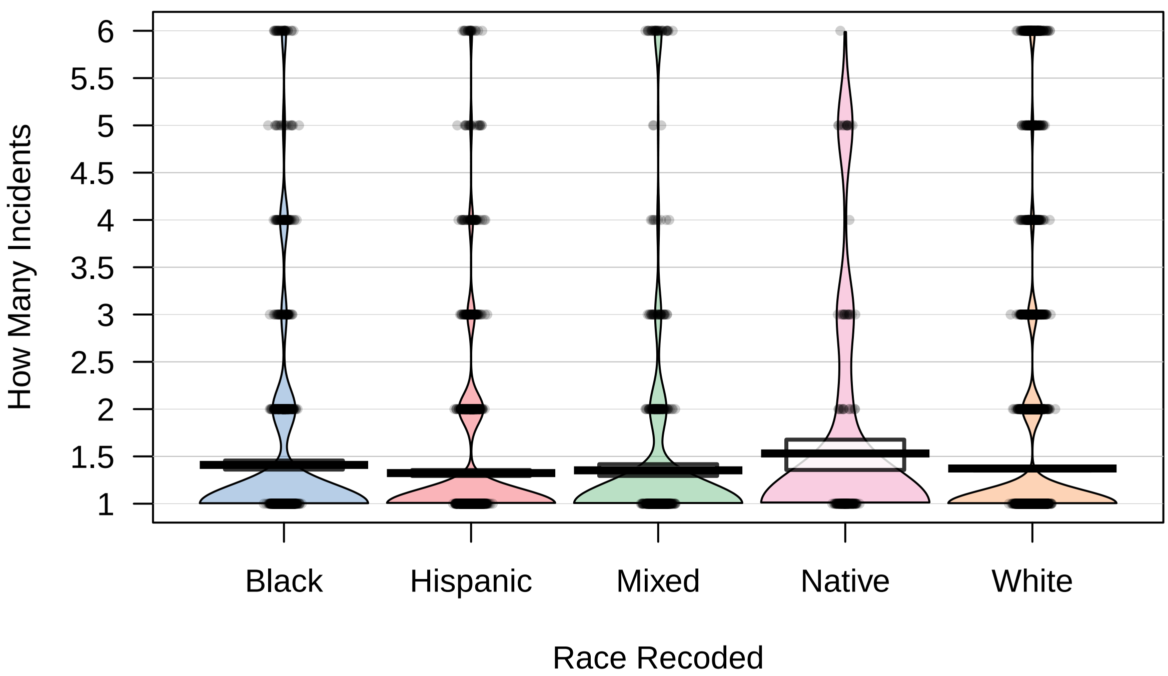



```
race_plot <-ggplot(dataset_rename,
                       aes(x = `How Many Incidents`,
                           fill = `Race Recoded`))+
  geom_bar(stat = "count",
           position = position_dodge()) +
  scale_fill_manual('Legend',
                    values = c('White' = 'dodgerblue', 'Black' = 'purple', 'Native' = 'red', 'Hispanic'
  scale_x_continuous(breaks = round(seq(min(0), max(6), by = 1),1)) +
  xlab("Number of Incidents by Race") +
  ylab("Frequency")

race_plot
```

```
## Warning: Removed 264395 rows containing non-finite values (`stat_count()`).
```




```
pirateplot(`How Many Incidents` ~ `Race Recoded` , data = dataset_rename)
```



Considering the great disparity between the representation of the many different races in our sample, we decided to focus on sex for the next part of our analysis since there is a more equal number of male and female respondents.

Density Concentration of Attempted Aggravated Battery ~ Sex

The attempted battery plots show the difference in the average intensity of attempted aggravated battery on victims between genders. Average intensity is defined as the mean value of attempted aggravated battery instances (which itself is an average of multiple values), rounded to three decimal places. That is, the intensity of victimisation - how likely is that a respondent was victim to one or more instances of attempted aggravated battery.

The bar plot shows that the average intensity is greater for males (.10) than for females (.09) This would mean that male victims experience attempted aggravated battery more intensely than women. However, this difference is not statistically significant. We checked the probability of superiority statistic, which tells us the percentage that someone picked at random from the Male group will have a higher average intensity of attempted aggravated battery than someone picked from the Female group. The probability of superiority was 0.52 with a 95% confidence interval ranging from [0.50, 0.55]. Because the confidence interval is so close to 0.50 (indicating a coin-flip chance) we cannot be confident that a victim of attempted aggravated battery of one gender is more likely to have a more intense victimization than the other gender. Essentially, it is a 50/50 chance that a victim that experiences a more intense attempted battery is of one gender over the other.

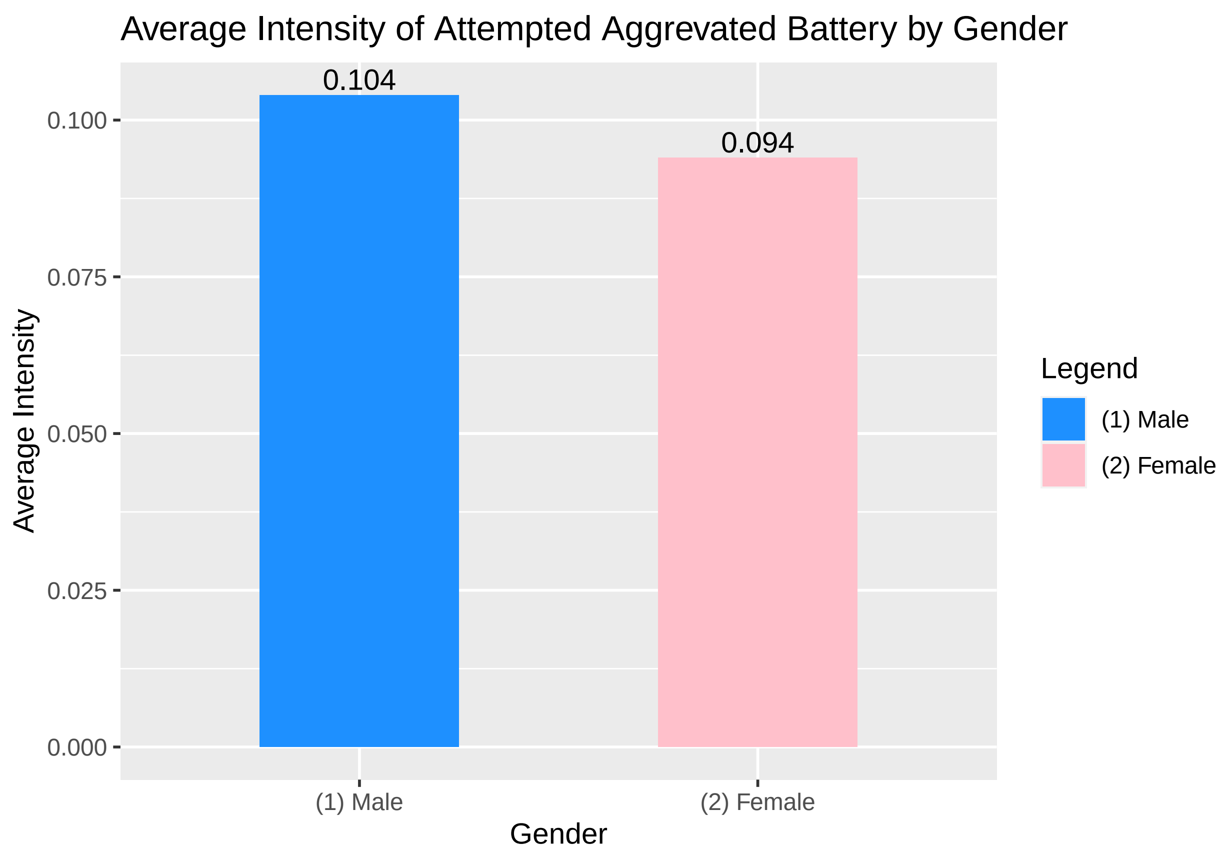
We can see the same uncertainty in the pirateplot(). This plot shows that the spread of the intensity of attempted battery is extremely similar between male victims and female victims. Both genders have the majority of their responses around 0 and 0.2 intensity, and both taper off equally as the intensity increases, all the way up to 0.6. Also, the error bars around the means of each gender overlap, which means we cannot say for certain that the mean of one gender's intensity of attempted aggravated battery victimization is greater than the other gender. Therefore, we cannot conclude that males experience a more intense attempted aggravated battery than females.

```
library(effectsize)
```

```
##
## Attaching package: 'effectsize'

## The following object is masked from 'package:psych':
##
##     phi
```

```
att_agg_batt_lay <-
  dataset_1%>%
  select(V3018, att_agg_batt_avg) %>%
  na.omit() %>%
  group_by(V3018) %>%
  summarize(att_agg_batt_avg = mean(att_agg_batt_avg))

att_agg_batt_lay$att_agg_batt_avg <- round(att_agg_batt_lay$att_agg_batt_avg, digits = 3)

ggplot(data = att_agg_batt_lay,
       aes(V3018, att_agg_batt_avg, fill = V3018)) +
  geom_bar(stat = 'identity', width = 0.5) +
  scale_fill_manual('Legend',
                    values = c('(1) Male' = 'dodgerblue', '(2) Female' = 'pink')) +
  geom_text(aes(label = att_agg_batt_avg),
            position = position_dodge(),
            vjust=-0.25) + xlab("Gender") +
  ylab("Average Intensity") +
  ggtitle("Average Intensity of Attempted Aggrevated Battery by Gender")
```

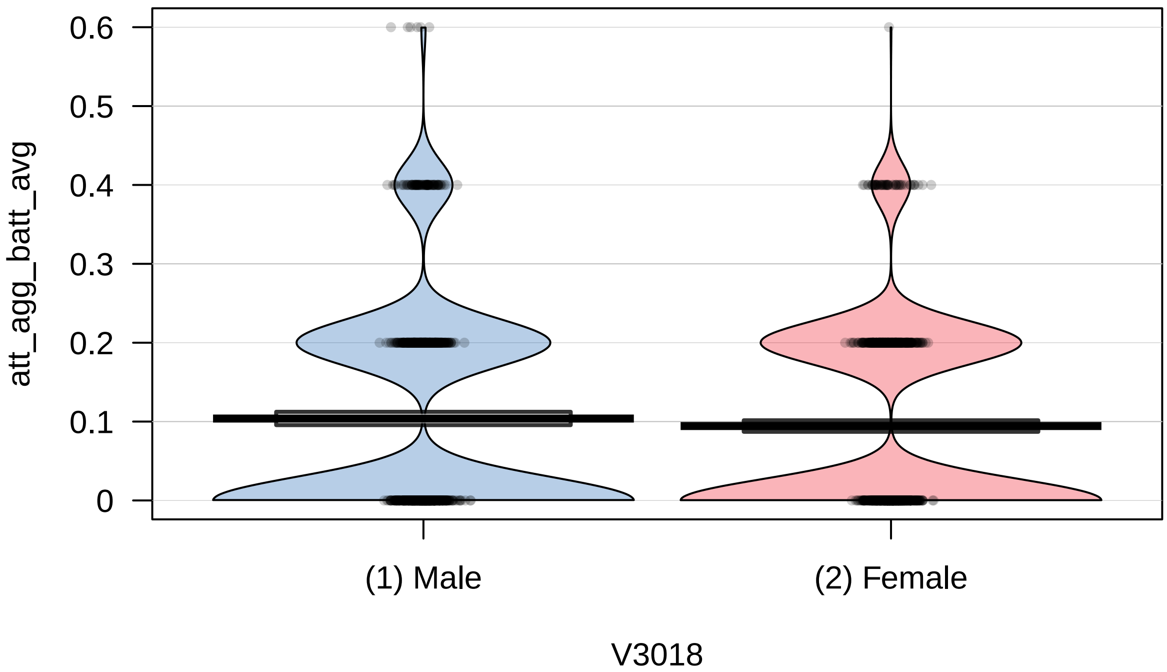
```
## Warning: Width not defined
## i Set with 'position_dodge(width = ...)'
```




```r
p_superiority(att_agg_batt_avg ~ V3018, data = dataset_1)
```

```
## Warning: Missing values detected. NAs dropped.

## Pr(superiority) |       95% CI
## ------------------------------
## 0.52            | [0.50, 0.55]
```

```r
#0.52, [0.50, 0.55]

pirateplot(att_agg_batt_avg ~ V3018, data = dataset_1)
```



Density Concentration of Attempted Battery ~ Sex

The attempted battery plots show the difference in the average intensity of attempted battery on victims between genders. Average intensity is defined as the mean value of attempted battery instances (which itself is an average of multiple values), rounded to three decimal places. That is, the intensity of victimisation - how likely is that a respondent was victim to one or more instances of attempted battery.

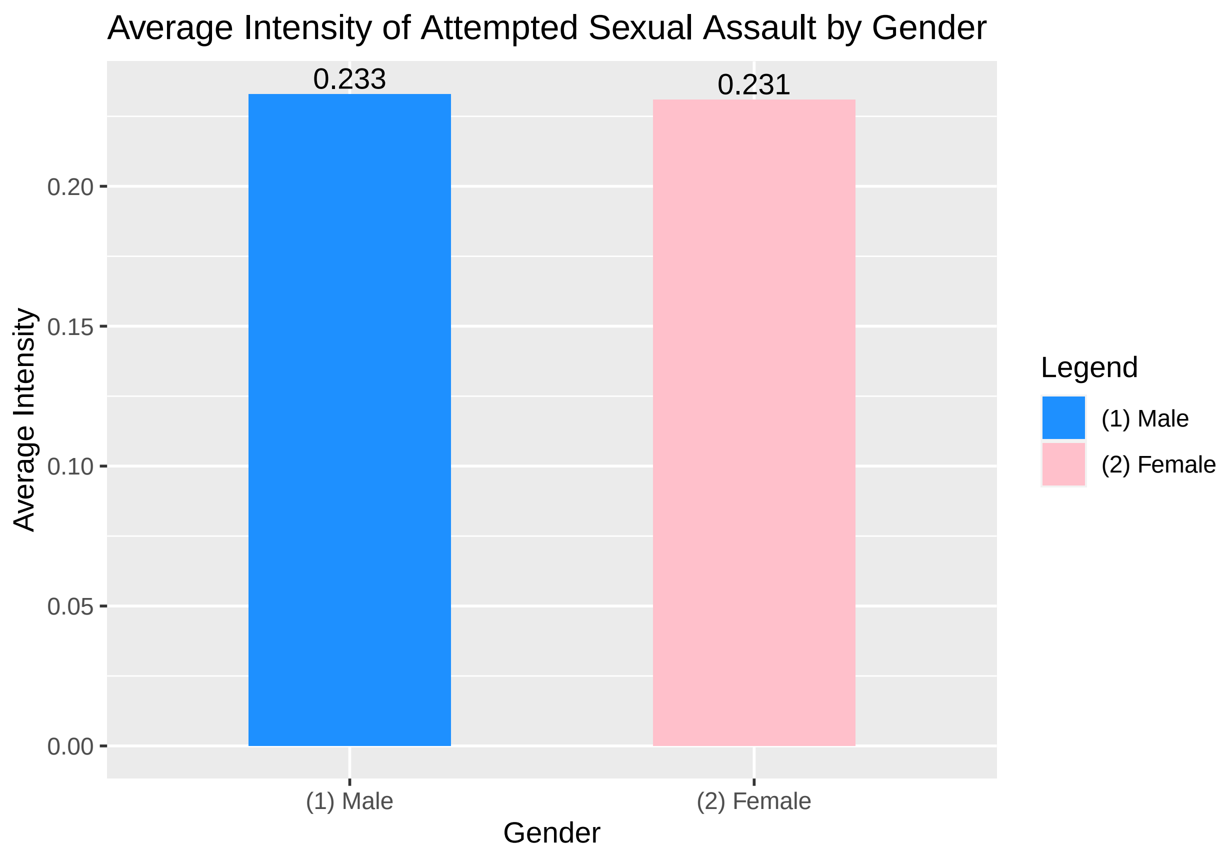
The bar plot shows that the average intensity is greater for males (.23) than for females (.231). This would mean that male victims experience attempted battery more intensely than women. However, this difference is not statistically significant. We checked the probability of superiority statistic, which tells us the percentage that someone picked at random from the Male group will have a higher average intensity of attempted battery than someone picked from the Female group. The probability of superiority was 0.50 with a 95% confidence interval ranging from [0.48, 0.53]. Because the confidence interval is so close to 0.50 (indicating a coin-flip chance) we cannot be confident that a victim of attempted battery of one gender is more likely to have a more intense victimization than the other gender.

We can see the same uncertainty in the pirateplot(). This plot shows that the spread of the intensity of attempted battery is extremely similar between male victims and female victims. Both genders have the majority of their responses around 0.2 and 0.4 intensity, and both taper off equally as the intensity increases, all the way up to 1. Also, the error bars around the means of each gender overlap, which means we cannot say for certain that the mean of one gender's intensity of attempted battery victimization is greater than the other gender. Therefore, we cannot conclude that males experience a more intense attempted battery than females.

```
att_batt_lay <-
  dataset_2 %>% select(V3018, att_batt_avg) %>%
  na.omit() %>%
  group_by(V3018) %>%
  summarize(att_battery_mean = mean(att_batt_avg))

att_batt_lay$att_battery_mean <- round(att_batt_lay$att_battery_mean, digits = 3)
```

```r
ggplot(data = att_batt_lay,
       aes(V3018, att_battery_mean, fill = V3018)) +
  geom_bar(stat = 'identity', width = 0.5) +
  scale_fill_manual('Legend',
                    values = c('(1) Male' = 'dodgerblue',
                               '(2) Female' = 'pink',
                               '(3) Transgender' = 'purple')) +
  geom_text(aes(label = att_battery_mean),
            position = 'dodge', vjust=-0.25) +
  xlab("Gender") +
  ylab("Average Intensity") +
  ggtitle("Average Intensity of Attempted Sexual Assault by Gender")
```

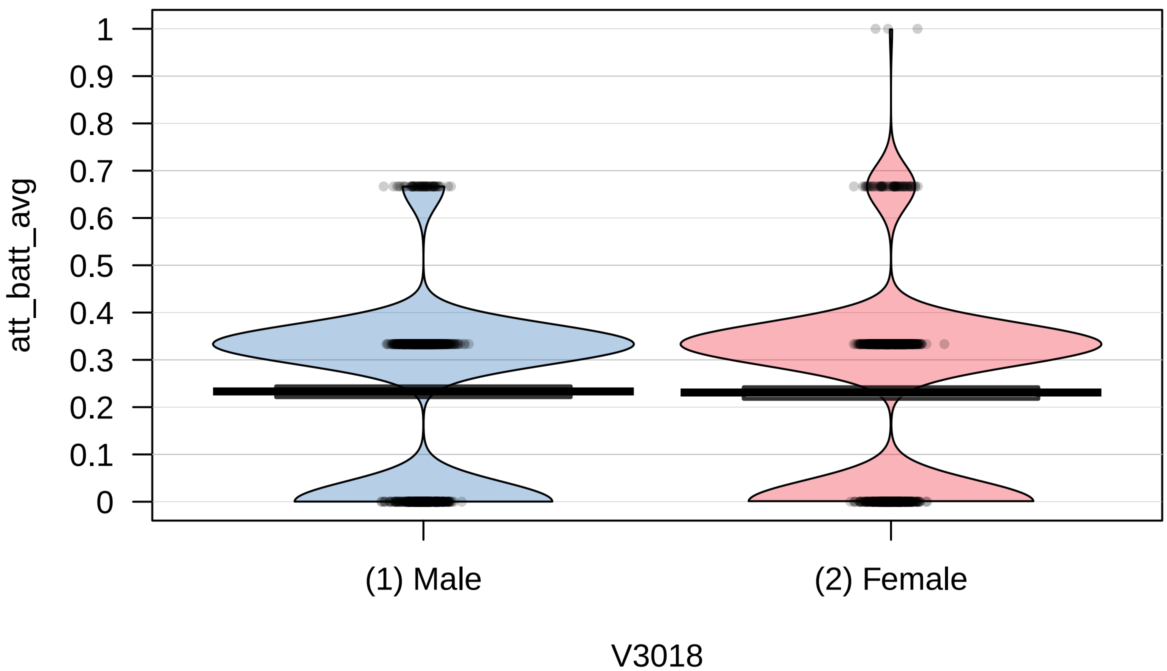
```
## Warning: Width not defined
## i Set with ‘position_dodge(width = ...)‘
```




```r
p_superiority(att_batt_avg ~ V3018, data = dataset_2)
```

```
## Warning: Missing values detected. NAs dropped.

## Pr(superiority) |       95% CI
## --------------------------------
## 0.50            | [0.48, 0.53]
```

```r
#0.50, [0.48, 0.53]

pirateplot(att_batt_avg ~ V3018, data = dataset_2)
```



Density Concentration of Attempted Sexual Assault ~ Sex

The attempted sexual assault plots show the difference in the average intensity of attempted battery on victims between genders.

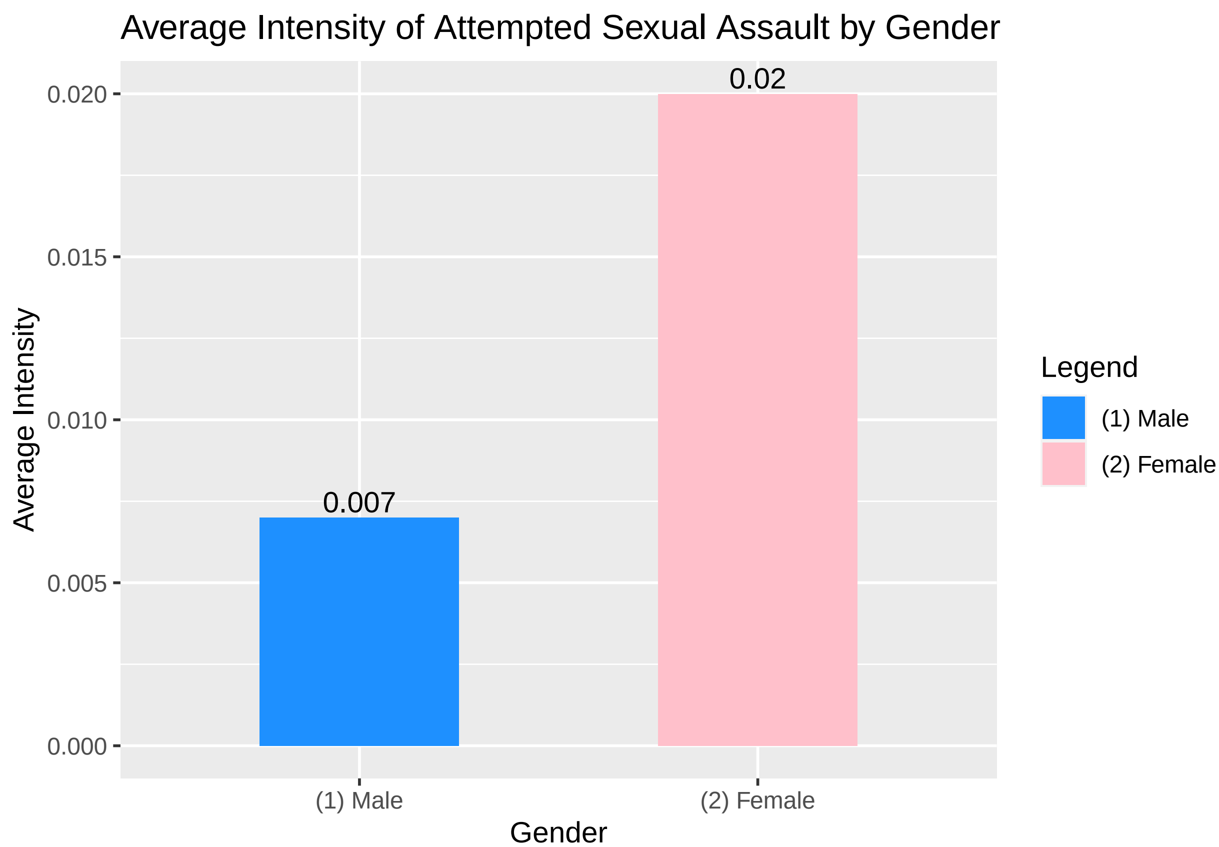
The bar plot shows that the average intensity is greater for females (.02) than for females (.007). This would mean that female victims experience attempted sexual assault more intensely than men However, this difference is not statistically significant. We checked the probability of superiority statistic, which tells us the percentage that someone picked at random from the Male group will have a higher average intensity of attempted battery than someone picked from the Female group. The probability of superiority was 0.45 with a 95% confidence interval ranging from [0.42, 0.47]. Because the confidence interval is so close to 0.50 (indicating a coin-flip chance) we cannot be confident that a victim of attempted battery of one gender is more likely to have a more intense victimization than the other gender.

We can see the same uncertainty in the pirateplot(). This plot shows that the spread of the intensity of attempted battery is extremely similar between male victims and female victims. Both genders have the majority of their responses around 0 and 0.04 intensity, and both taper off equally as the intensity increases, up to .36 - all of which are minuscule values. Also, the error bars around the means of each gender overlap, which means we cannot say for certain that the mean of one gender's intensity of attempted sexual assault victimization is greater than the other gender. Therefore, we cannot conclude that females experience a more intense attempted sexual assault than females.

```r
fem_SA_lay <- dataset_3 %>%
  select(V3018, att_SA_avg) %>%
  na.omit() %>%
  group_by(V3018) %>%
  summarize(SA_mean = mean(att_SA_avg))
```

```r
fem_SA_lay$SA_mean <- round(fem_SA_lay$SA_mean, digits = 3)

ggplot(data = fem_SA_lay,
       aes(V3018, SA_mean, fill = V3018)) +
  geom_bar(stat = 'identity', width = 0.5) +
  scale_fill_manual('Legend',
                    values = c('(1) Male' = 'dodgerblue', '(2) Female' = 'pink')) +
  geom_text(aes(label = SA_mean),
            position = 'dodge', vjust=-0.25) +
  xlab("Gender") +
  ylab("Average Intensity") +
  ggtitle("Average Intensity of Attempted Sexual Assault by Gender")
```

```
## Warning: Width not defined
## i Set with `position_dodge(width = ...)`
```

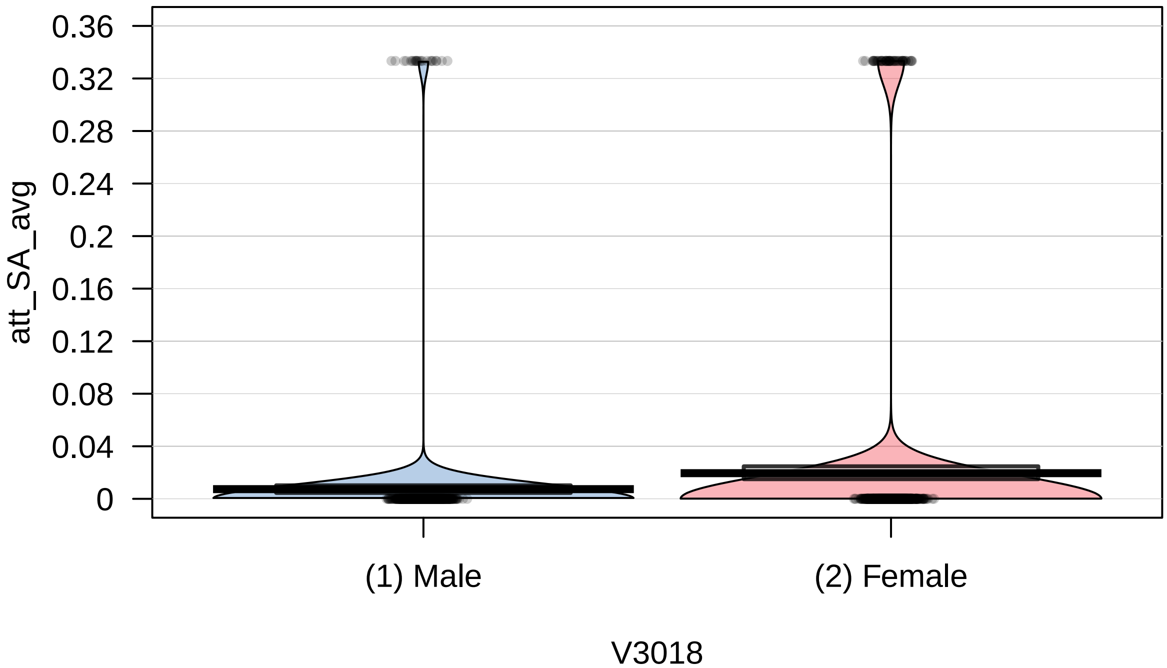



```r
p_superiority(att_SA_avg ~ as_factor(V3018), data = dataset_3)
```

```
## Warning: 'y' is numeric but has only 2 unique values.
##   If this is a grouping variable, convert it to a factor.

## Warning: Missing values detected. NAs dropped.

## Warning: 'y' is numeric but has only 2 unique values.
##   If this is a grouping variable, convert it to a factor.
```

```
## Pr(superiority) |       95% CI
## ------------------------------
## 0.45             | [0.42, 0.47]
```

```r
#.45 [.42, .47]

pirateplot(att_SA_avg ~ V3018, data = dataset_3)
```



Density Concentration of Aggravated Battery ~ Sex

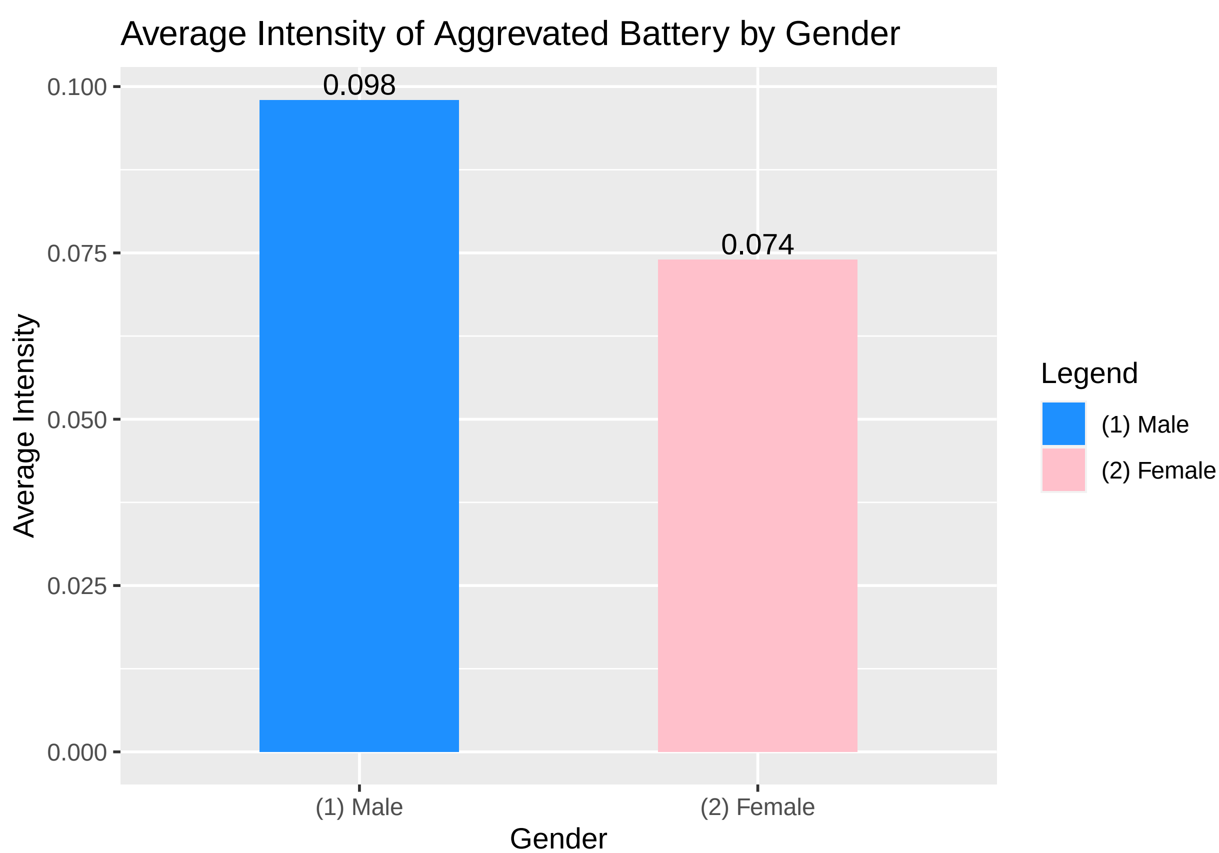
The aggravated battery plots show the difference in the average intensity of aggravated battery on victims between genders.

The bar plot shows that the average intensity is greater for males (.09) than for females (.07) This would mean that male victims experience aggravated battery more intensely than women. However, this difference is not statistically significant. We checked the probability of superiority statistic, which tells us the percentage that someone picked at random from the Male group will have a higher average intensity of aggravated battery than someone picked from the Female group. The probability of superiority was 0.56 with a 95% confidence interval ranging from [0.53, 0.60]. Because the confidence interval is so close to 0.50 (indicating a coin-flip chance) we cannot be confident that a victim of aggravated battery of one gender is more likely to have a more intense victimization than the other gender.

We can see the same uncertainty in the pirateplot(). This plot shows that the spread of the intensity of aggravated battery is extremely similar between male victims and female victims. Both genders have the majority of their responses around 0 and 0.24 intensity, and both taper off equally as the intensity increases, up to 0.52. Also, the error bars around the means of each gender overlap, which means we cannot say for certain that the mean of one gender's intensity of aggravated battery victimization is greater than the other gender. Therefore, we cannot conclude that males experience a more intense aggravated battery than females.

```r
agg_batt_lay <- dataset_4 %>%
  select(V3018, agg_bat_avg) %>%
  na.omit() %>% group_by(V3018) %>%
  summarize(agg_batt_avg = mean(agg_bat_avg))

agg_batt_lay$agg_batt_avg <- round(agg_batt_lay$agg_batt_avg, digits = 3)

ggplot(data = agg_batt_lay,
       aes(V3018, agg_batt_avg, fill = V3018)) +
  geom_bar(stat = 'identity', width = 0.5) +
  scale_fill_manual('Legend',
                    values = c('(1) Male' = 'dodgerblue',
                               '(2) Female' = 'pink')) +
  geom_text(aes(label = agg_batt_avg),
            position = position_dodge(),
            vjust=-0.25) +
  xlab("Gender") +
  ylab("Average Intensity") +
  ggtitle("Average Intensity of Aggrevated Battery by Gender")
```

```
## Warning: Width not defined
## i Set with ‘position_dodge(width = ...)‘
```

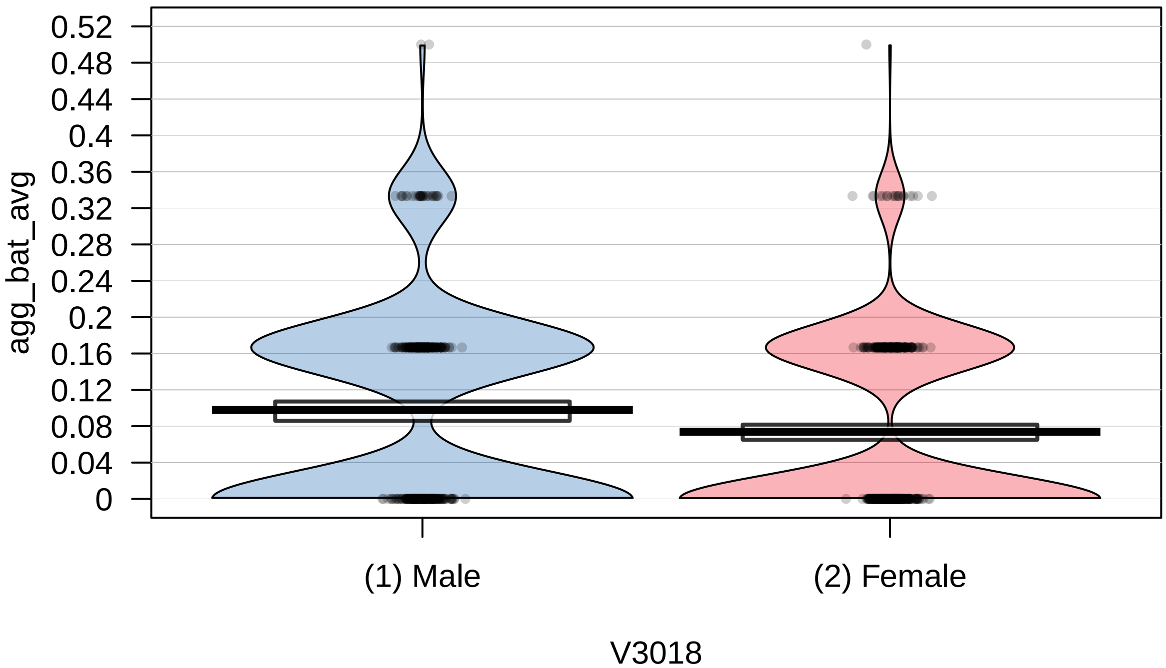



```r
p_superiority(agg_bat_avg ~ V3018, data = dataset_4)
```

```
## Warning: Missing values detected. NAs dropped.
```

```
## Pr(superiority) |       95% CI
## ------------------------------
## 0.56             | [0.53, 0.60]
```

```
#.56 [.53, .60]

pirateplot(agg_bat_avg ~ V3018, data = dataset_4)
```



Density Concentration of Battery ~ Sex

The battery plots show the difference in the average intensity of battery on victims between genders.

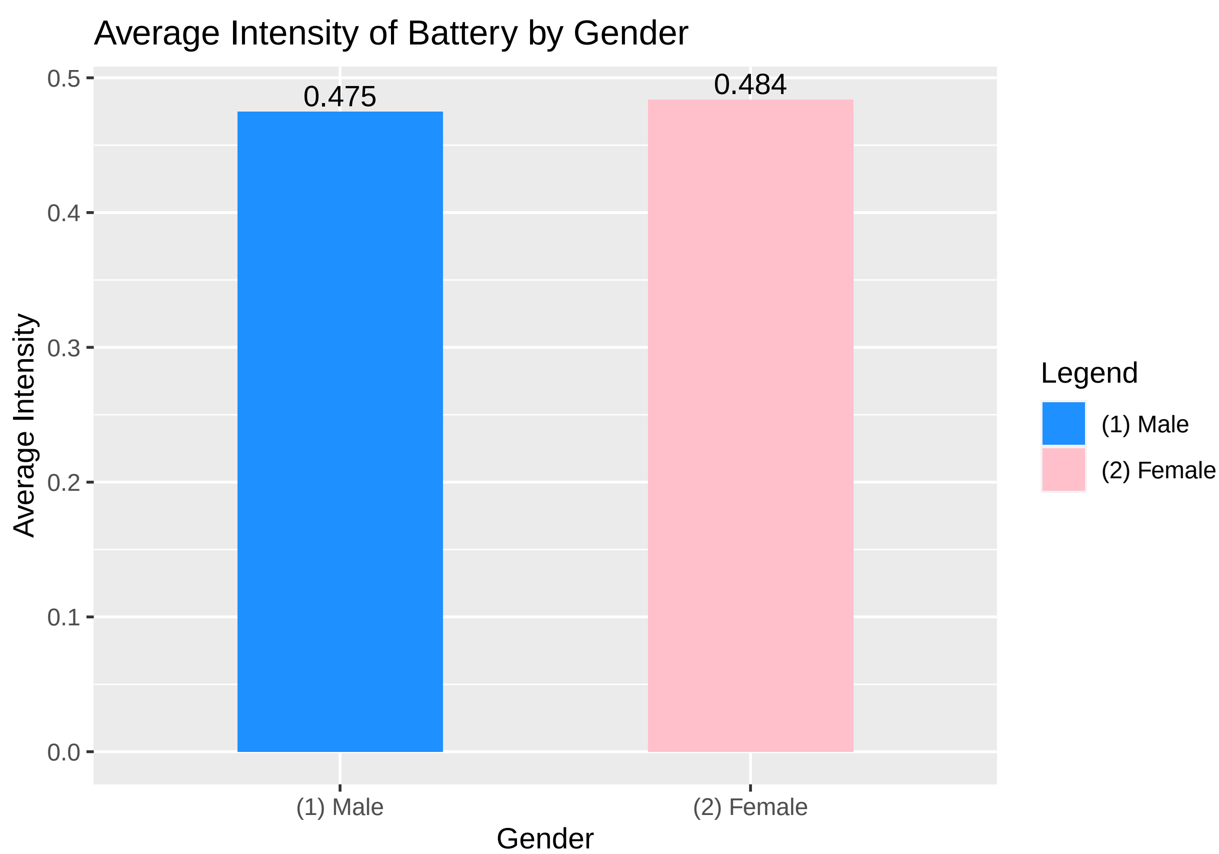
The bar plot shows that the average intensity is lesser for males (.475) than for females (.484) This would mean that female victims experience battery more intensely than women. However, this difference is not statistically significant. We checked the probability of superiority statistic, which tells us the percentage that someone picked at random from the male group will have a higher average intensity of battery than someone picked from the Female group. The probability of superiority was 0.49 with a 95% confidence interval ranging from [0.46, 0.53]. Because the confidence interval is so close to 0.50 (indicating a coin-flip chance) we cannot be confident that a victim of battery of one gender is more likely to have a more intense victimization than the other gender.

We can see the same uncertainty in the pirateplot(). This plot shows that the spread of the intensity of battery is extremely similar between male victims and female victims. Both genders have the majority of their responses around 0.4 and 0.7 intensity, and both taper off equally as the intensity increases, up to 1. Also, the error bars around the means of each gender overlap, which means we cannot say for certain that the mean of one gender's intensity of battery victimization is greater than the other gender. Therefore, we cannot conclude that females experience a more intense battery than females.

```
batt_lay <- dataset_5 %>%
  select(V3018, bat_avg) %>%
  na.omit() %>%
  group_by(V3018) %>%
```

```r
  summarize(batt_avg = mean(bat_avg))

batt_lay$batt_avg <- round(batt_lay$batt_avg, digits = 3)

ggplot(data = batt_lay,
       aes(V3018, batt_avg, fill = V3018)) +
  geom_bar(stat = 'identity', width = 0.5) +
  scale_fill_manual('Legend',
                    values = c('(1) Male' = 'dodgerblue',
                               '(2) Female' = 'pink')) +
  geom_text(aes(label = batt_avg),
            position = position_dodge(),
            vjust=-0.25) +
  xlab("Gender") +
  ylab("Average Intensity") +
  ggtitle("Average Intensity of Battery by Gender")
```

```
## Warning: Width not defined
## i Set with `position_dodge(width = ...)`
```

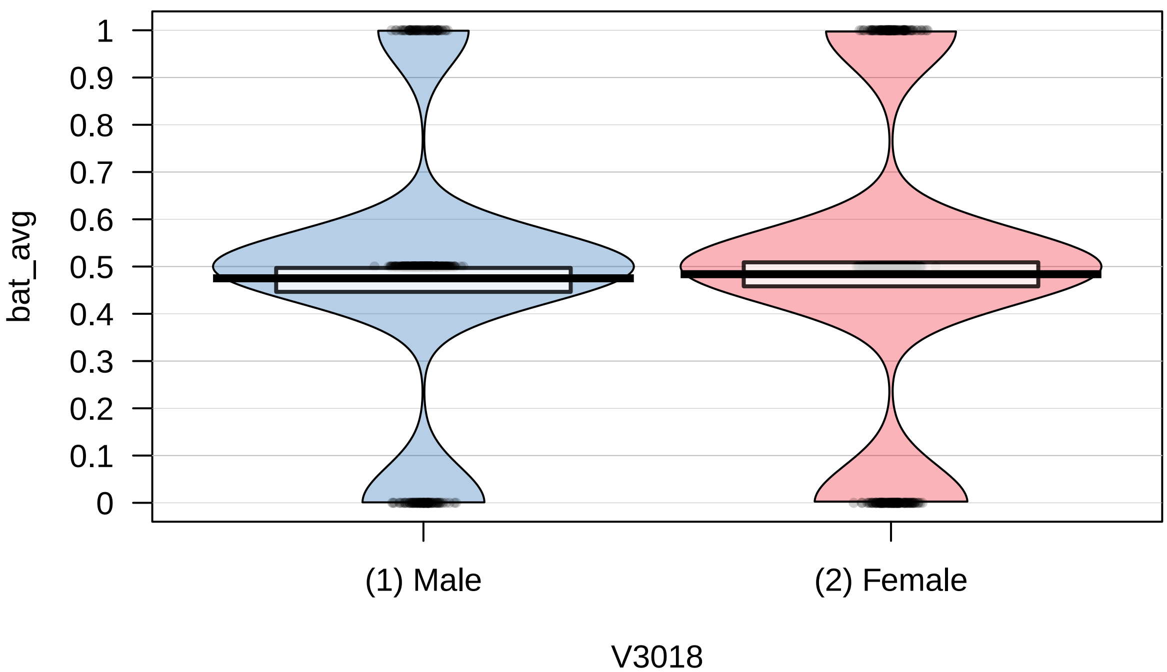



```r
p_superiority(bat_avg ~ V3018, data = dataset_5)
```

```
## Warning: Missing values detected. NAs dropped.

## Pr(superiority) |       95% CI
```

```
## ------------------------------
## 0.49            | [0.46, 0.53]
```

```r
#.49 [.46, .53]

pirateplot(bat_avg ~ V3018, data = dataset_5)
```



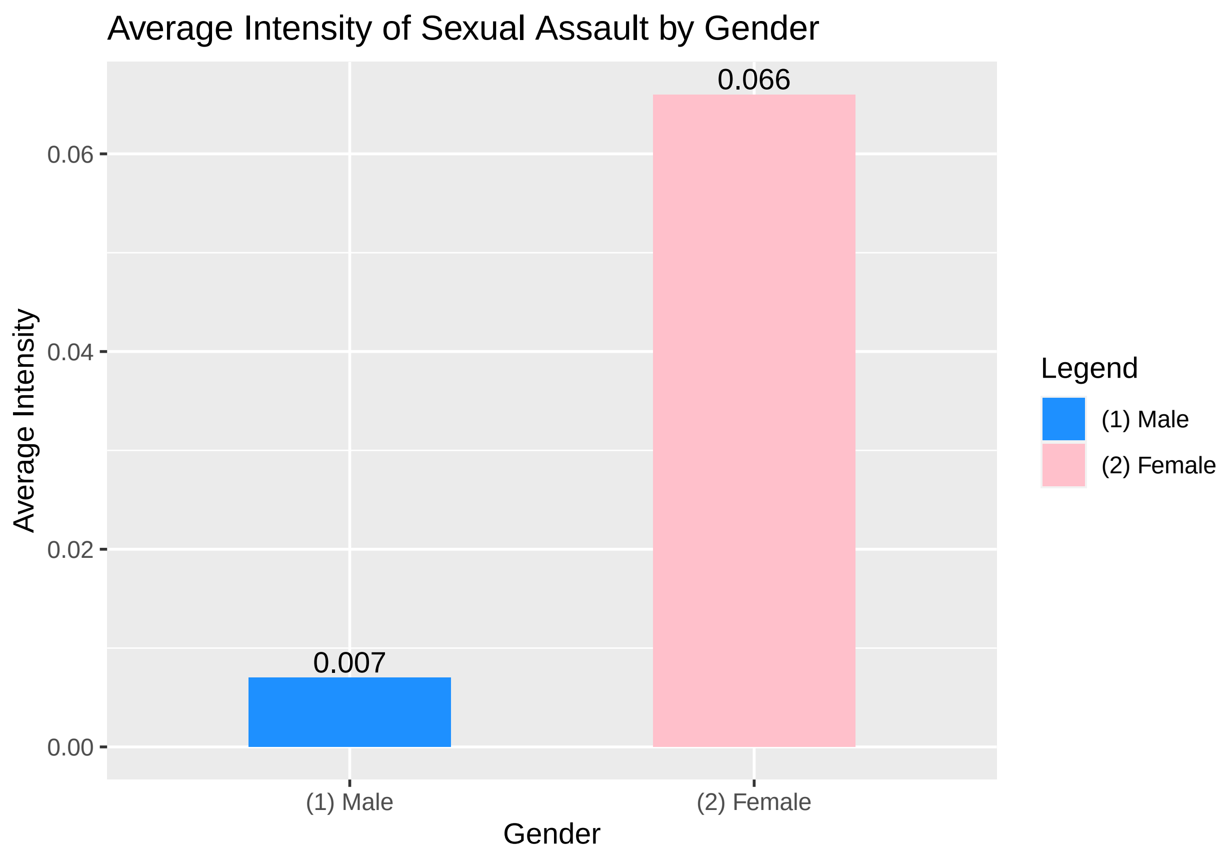
Density Concentration of Sexual Assault ~ Sex

The sexual assault plots show a different story than the rest of the variables. The bar graph shows the intensity of sexual assault experiences by victims separated by gender. It shows that women experience a much greater intensity of sexual assault than men on average. Unlike with the other variables, the probability of superiority statistic reinforces this divide. With men as the reference, the statistic was 0.35 with a [0.32, 0.38] 95% confidence interval. This means that there is only a 35% chance that a male experiences sexual assault at a greater intensity than females, which means there is a 75% chance that a female experiences sexual assault at a higher intensity. The spread of the data also supports the difference between men and women in intensity of sexual assault. The pirateplot() shows that the spread of the data for women goes much higher than for men. The spread of the data for women goes all the way up to around 0.65, while the spread of data for men only goes up to 0.35. Also, women have more data clustered around the 0.35 intensity than men do. The vast majority of the data for the intensity of sexual assault for men is near 0. Also, the plot shows the average intensity of sexual assault for women looks to be around 0.7, while the average for men seems to be around 0.1. Since the error bars on these averages do not overlap, we can say that women experience a more intense sexual assault than men on average.

```r
sa_lay <- dataset_6 %>%
  select(V3018, SA_avg) %>%
  na.omit() %>%
  group_by(V3018) %>%
  summarize(sa_avg = mean(SA_avg))

sa_lay$sa_avg <- round(sa_lay$sa_avg, digits = 3)
```

```r
ggplot(data = sa_lay,
       aes(V3018, sa_avg, fill = V3018)) +
  geom_bar(stat = 'identity', width = 0.5) +
  scale_fill_manual('Legend', values = c('(1) Male' = 'dodgerblue',
                                         '(2) Female' = 'pink')) +
  geom_text(aes(label = sa_avg),
            position = position_dodge(),
            vjust=-0.25) +
  xlab("Gender") +
  ylab("Average Intensity") +
  ggtitle("Average Intensity of Sexual Assault by Gender")
```

```
## Warning: Width not defined
## i Set with ‘position_dodge(width = ...)‘
```

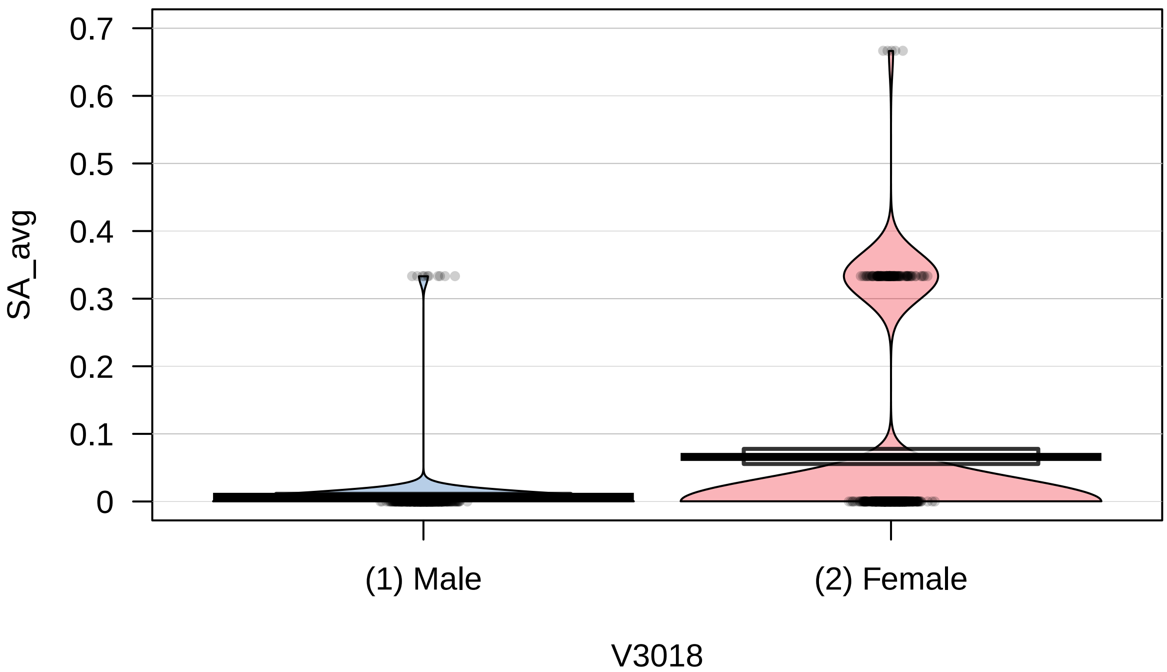



```r
p_superiority(SA_avg ~ V3018, data = dataset_6)
```

```
## Warning: Missing values detected. NAs dropped.
```

```
## Pr(superiority) |       95% CI
## ------------------------------
## 0.35            | [0.32, 0.38]
```

```r
#.35 [.32, .38]

pirateplot(SA_avg ~ V3018, data = dataset_6)
```

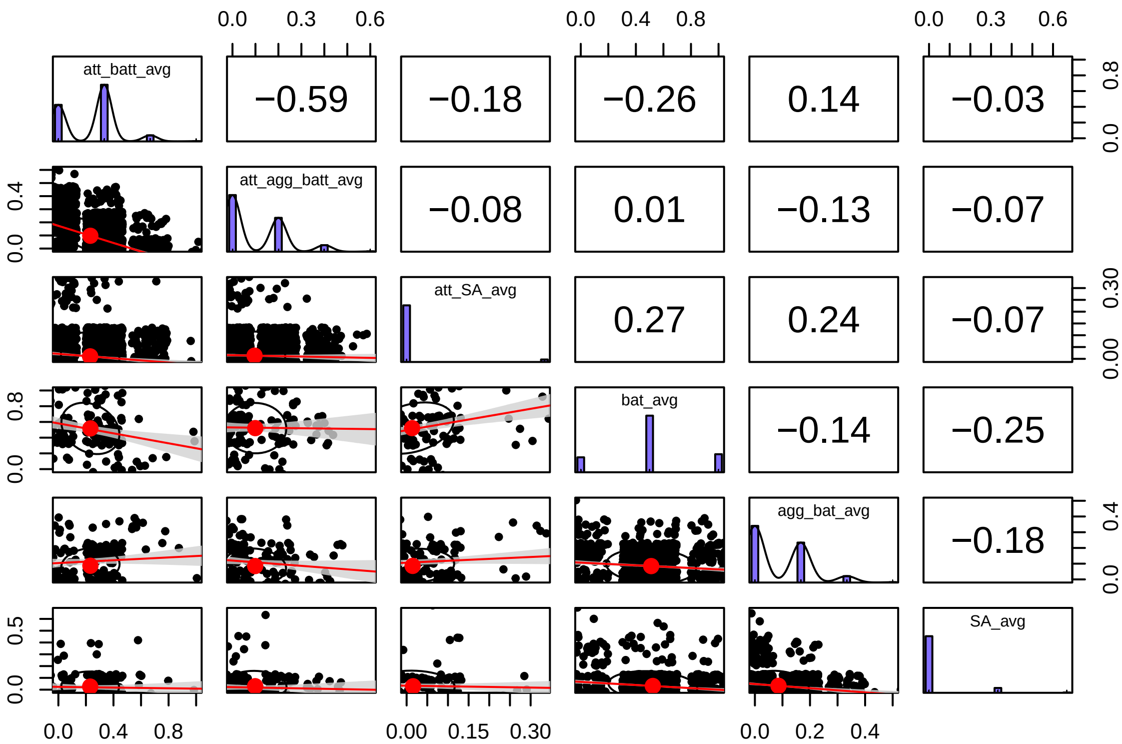


Multiple Regression

After creating a visual representation of the difference in instances of violent personal crime between genders, we wanted to learn *what is the correlation between attempts of violent crime and actual occurrence of violent crime*. To learn this, we conducted a multiple regression analysis between each variable of actual violent crime and the cumulative variable showing attempted violent crime and found some interesting results. We found that Battery and Aggravated Battery were both positively correlated with attempts of violent personal crime, and their highest unique predictor was attempted sexual assault. However, actual sexual assault was negatively correlated with all attempts of violent personal crime, with the strongest predictor also being attempted sexual assault.

```r
efa_data_1 <- dataset_6A %>%
  select(IDPER,
         att_batt_avg,
         att_agg_batt_avg,
         att_SA_avg) %>%
  na.omit(efa_data_1)

efa_data_2 <- dataset_6A %>%
  select(IDPER,
         bat_avg,
         agg_bat_avg,
         SA_avg) %>%
  na.omit(efa_data_2)

efa_data <- full_join(efa_data_1, efa_data_2, join_by(IDPER), relationship = "many-to-many")
```

```r
efa_data_final <- efa_data %>%
  select(-IDPER)

library(lm.beta)

pairs.panels(efa_data_final,
             jiggle = TRUE,
             lm = TRUE,
             ci = TRUE,
             pch = 20,
             #scale = TRUE,
             hist.col = "slateblue1",
             na.rm = TRUE)
```




```r
multi_reg_1 <- lm(bat_avg ~ att_batt_avg + att_agg_batt_avg + att_SA_avg,
                  data = efa_data_final)
summary(multi_reg_1)
```

```
##
## Call:
## lm(formula = bat_avg ~ att_batt_avg + att_agg_batt_avg + att_SA_avg,
##     data = efa_data_final)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.55801 -0.05801  0.01054  0.05004  0.61859
##
## Coefficients:
```

```
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        0.59751    0.05885  10.153   <2e-16 ***
## att_batt_avg      -0.32416    0.13446  -2.411    0.017 *
## att_agg_batt_avg -0.19750    0.23757  -0.831    0.407
## att_SA_avg         0.63055    0.30459   2.070    0.040 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2983 on 162 degrees of freedom
##   (2699 observations deleted due to missingness)
## Multiple R-squared:  0.1109, Adjusted R-squared:  0.09444
## F-statistic: 6.736 on 3 and 162 DF,  p-value: 0.0002606
```

```r
#R2 = .1109, highest unique att_SA_avg (.631)

multi_reg_2 <- lm(agg_bat_avg ~ att_batt_avg + att_agg_batt_avg + att_SA_avg,
                data = efa_data_final)
summary(multi_reg_2)
```

```
##
## Call:
## lm(formula = agg_bat_avg ~ att_batt_avg + att_agg_batt_avg +
##     att_SA_avg, data = efa_data_final)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.20513 -0.10413  0.04580  0.06254  0.26013
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        0.05647    0.02133   2.648  0.00891 **
## att_batt_avg       0.14300    0.04872   2.935  0.00382 **
## att_agg_batt_avg  0.08369    0.08609   0.972  0.33244
## att_SA_avg         0.44599    0.11037   4.041 8.21e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1081 on 162 degrees of freedom
##   (2699 observations deleted due to missingness)
## Multiple R-squared:  0.1125, Adjusted R-squared:  0.0961
## F-statistic: 6.847 on 3 and 162 DF,  p-value: 0.0002261
```

```r
#R2 = .1125, highest unique - att_SA_avg (.446)

multi_reg_3 <- lm(SA_avg ~ att_batt_avg + att_agg_batt_avg + att_SA_avg,
                data = efa_data_final)
summary(multi_reg_3)
```

```
##
## Call:
## lm(formula = SA_avg ~ att_batt_avg + att_agg_batt_avg + att_SA_avg,
##     data = efa_data_final)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.03585 -0.03585 -0.03177 -0.00051  0.63489
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)       0.06303    0.01839   3.428 0.000771 ***
## att_batt_avg     -0.08154    0.04201  -1.941 0.054007 .
## att_agg_batt_avg -0.15631    0.07423  -2.106 0.036766 *
## att_SA_avg       -0.18910    0.09517  -1.987 0.048603 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0932 on 162 degrees of freedom
##   (2699 observations deleted due to missingness)
## Multiple R-squared:  0.03564,    Adjusted R-squared:  0.01778
## F-statistic: 1.996 on 3 and 162 DF,  p-value: 0.1167
```

```
#R2 = .03564, highest unique = att_SA_avg (negatively correlated)
```

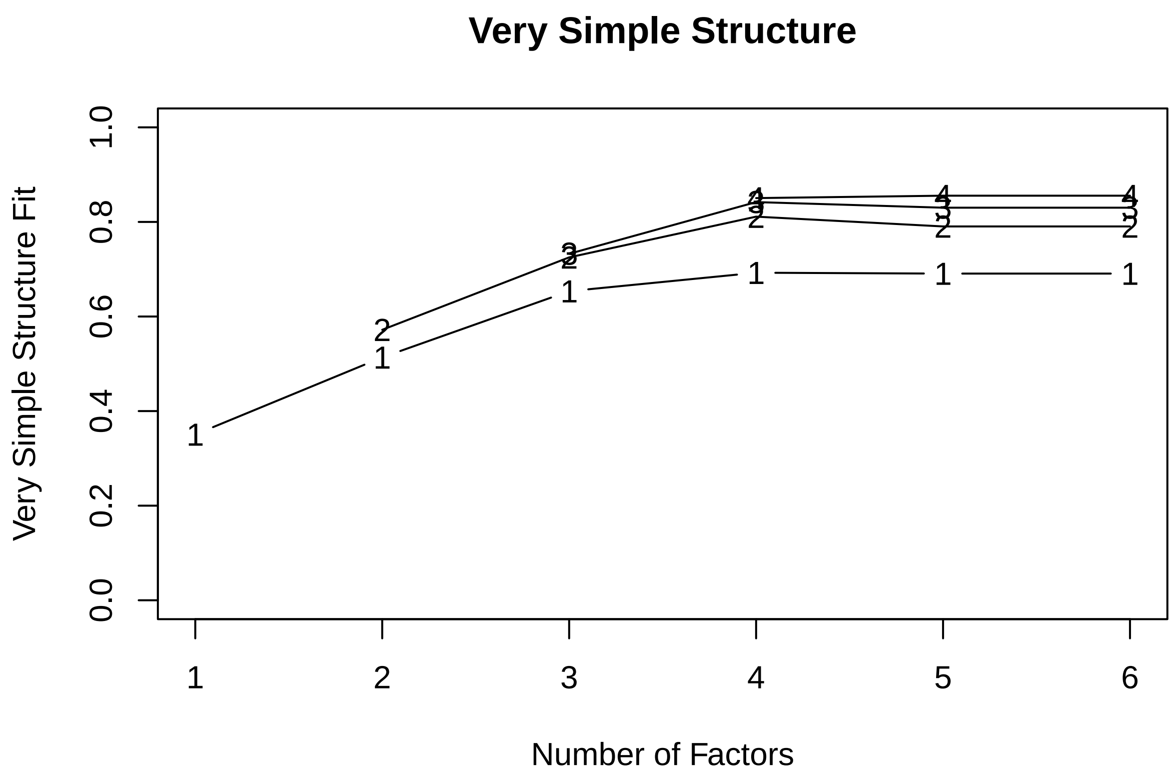
Exploratory Factor Analysis (EFA)

We were interested in learning whether these results were actually reflective of a systemic issue in the way criminal activity is reported, measured and charged in the field of criminal justice or if the results were indicative of a procedural error in our research. To explore this further, we conducted an exploratory factor analysis to check if our proposed model of attempted versus actual crime was the best way to evaluate the variables of interest.

The vss function showed no inflection point in MAP values, which was our initial indicator that the results were better attributed to error on our part. Similarly, the scree plot recommended a 0 factor model, which tells us that our data compilation was likely incorrect. Even so, we compared our 2 factor model (attempts vs. actual) with a 1 factor model (all variables categorized as personal violent crime). The RMSEA for the 1 factor model was .212 90% CI [.202, .233] and the RMSEA for 2 factor model was .231 90% CI [.216, .246]. Since our standard for adequate fit of model is RMSEA = .05 or below, neither of these models are a good fit. When we compared 90% CIs to compare goodness of fit between the two models, we computed .233-.202 = .31*2 = .62 and .246-.216 = .40, which was not <.62, therefore 1 factor model is not a reliably better fit than 2 factor. Furthermore, it is important to note that nfactors > 2 did not yield any result at all.

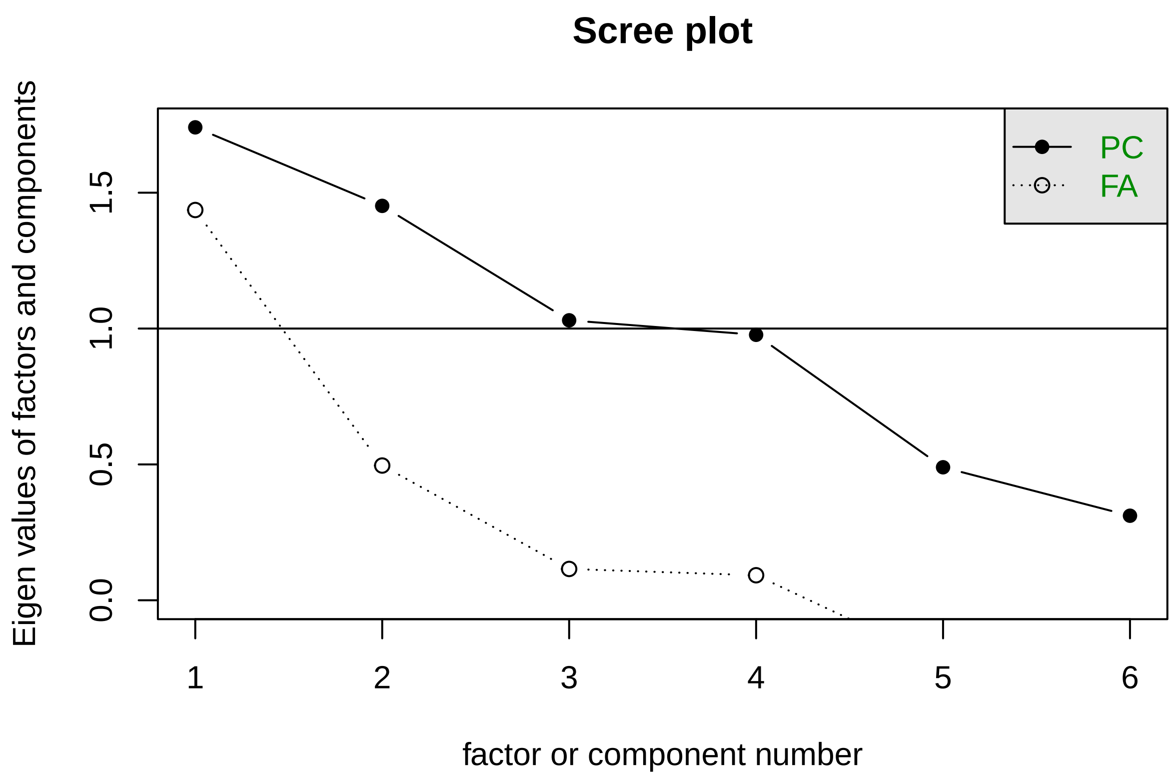
At this point, we thought these results probably occurred because we lost a lot of observations during our data cleaning process, and according to Revelle, since all our variables had communalities <.7 (except attempted battery), we needed greater number of subjects for our reliability model than what we had remaining after dropping NAs.

```
vss(efa_data_final)
```




```
##
## Very Simple Structure
## Call: vss(x = efa_data_final)
## VSS complexity 1 achieves a maximimum of 0.69  with  4  factors
## VSS complexity 2 achieves a maximimum of 0.81  with  4  factors
##
## The Velicer MAP achieves a minimum of 0.08  with  1  factors
## BIC achieves a minimum of  582.27  with  2  factors
## Sample Size adjusted BIC achieves a minimum of  594.97  with  2  factors
##
## Statistics by number of factors
##   vss1 vss2   map dof   chisq     prob sqresid  fit RMSEA  BIC SABIC complex
## 1 0.35 0.00 0.084   9 1.2e+03 2.5e-246     4.9 0.35  0.21 1099  1128     1.0
## 2 0.51 0.57 0.140   4 6.1e+02 1.4e-131     3.2 0.57  0.23  582   595     1.3
## 3 0.65 0.73 0.228   0 2.3e+02       NA     2.0 0.73    NA   NA    NA     1.3
## 4 0.69 0.81 0.434  -3 1.9e-10       NA     1.1 0.85    NA   NA    NA     1.4
## 5 0.69 0.79 1.000  -5 0.0e+00       NA     1.1 0.86    NA   NA    NA     1.5
## 6 0.69 0.79    NA  -6 0.0e+00       NA     1.1 0.86    NA   NA    NA     1.5
##    eChisq    SRMR eCRMS eBIC
## 1 1.6e+03 1.4e-01  0.18 1541
## 2 6.3e+02 8.5e-02  0.17  596
## 3 1.6e+02 4.3e-02    NA   NA
## 4 6.3e-11 2.7e-08    NA   NA
## 5 2.6e-18 5.5e-12    NA   NA
## 6 2.6e-18 5.5e-12    NA   NA
```

```
scree(efa_data_final)
```




```r
fa(efa_data_final,
   nfactors = 2,
   fm = "minres",
   rotate = "oblimin")
```

```
## Factor Analysis using method =  minres
## Call: fa(r = efa_data_final, nfactors = 2, rotate = "oblimin", fm = "minres")
## Standardized loadings (pattern matrix) based upon correlation matrix
##                    MR1   MR2   h2     u2 com
## att_batt_avg      0.99 -0.05 1.00 -0.003 1.0
## att_agg_batt_avg -0.61 -0.19 0.37  0.626 1.2
## att_SA_avg       -0.02  0.74 0.55  0.450 1.0
## bat_avg          -0.19  0.33 0.16  0.837 1.6
## agg_bat_avg       0.22  0.29 0.11  0.888 1.8
## SA_avg           -0.01 -0.22 0.05  0.950 1.0
##
##                        MR1  MR2
## SS loadings           1.43 0.82
## Proportion Var        0.24 0.14
## Cumulative Var        0.24 0.38
## Proportion Explained  0.64 0.36
## Cumulative Proportion 0.64 1.00
##
##  With factor correlations of
##       MR1   MR2
## MR1  1.00 -0.15
## MR2 -0.15  1.00
##
## Mean item complexity =  1.3
```

```
## Test of the hypothesis that 2 factors are sufficient.
##
## The degrees of freedom for the null model are  15  and the objective function was  0.95 with Chi Squa
## The degrees of freedom for the model are 4  and the objective function was  0.21
##
## The root mean square of the residuals (RMSR) is  0.13
## The df corrected root mean square of the residuals is  0.26
##
## The harmonic number of observations is  294 with the empirical chi square  156.67  with prob <  7.6e
## The total number of observations was  2865  with Likelihood Chi Square =  614.11  with prob <  1.4e-
##
## Tucker Lewis Index of factoring reliability =  0.152
## RMSEA index =  0.231  and the 90 % confidence intervals are  0.216 0.246
## BIC =  582.27
## Fit based upon off diagonal values = 0.92
```

```
#RMSEA = .231 [.216, .246]

incident_efa <- fa(efa_data_final,
   nfactors = 1,
   fm = "minres",
   rotate = "oblimin")

#RMSEA = .212, [.202, .233]

#for nfactors > 2, RMSEA not given.
#RMSEA went in the opposite direction to the expectation

incident_efa$communalities
```

```
##     att_batt_avg att_agg_batt_avg       att_SA_avg          bat_avg
##      0.995000000      0.299915466      0.015188476      0.058875763
##      agg_bat_avg           SA_avg
##      0.021586153      0.001049947
```

```
#Sample size not large enough for most variables except attempted battery
#Potentially because we dropped NA
```

Chronbach's Alpha

To check this, we used Cronbach's alpha to measure internal consistency within our two factors. The alpha function indicated that our attempted SA variable needed to be reverse scored, which may have impacted our multiple regression conclusions that actual violent crime was best predicted by attempted sexual assault, when in fact it was likely the worst predictor. In the attempted violent crime factor, if attempted SA was dropped, std.r value for chronbach's alpha increased from .48 to .75, further confirming that attempted SA was coded as a separate type of crime, separate from battery and aggravated battery. Furthermore, we found that attempted (std.r = .89) and actual instances (std.r = .89) of battery were the most reliable variables in attempted and actual violent crime factors respectively. This is likely due to the higher number of instances of battery compared to aggravated or sexual assault.

```r
alpha_attempts <- efa_data_1 %>%
  select(!IDPER)

alpha(alpha_attempts)
```

```
## Some items ( att_batt_avg ) were negatively correlated with the total scale and
## probably should be reversed.
## To do this, run the function again with the 'check.keys=TRUE' option

##
## Reliability analysis
## Call: alpha(x = alpha_attempts)
##
##   raw_alpha std.alpha G6(smc) average_r   S/N  ase mean    sd median_r
##       -2.2      -1.9   -0.61     -0.28 -0.65 0.12 0.11 0.051    -0.17
##
##  lower alpha upper     95% confidence boundaries
## -2.45 -2.22 -2
##
##  Reliability if an item is dropped:
##                 raw_alpha std.alpha G6(smc) average_r   S/N alpha se var.r
## att_batt_avg        -0.12     -0.15  -0.068    -0.068 -0.13    0.040    NA
## att_agg_batt_avg    -0.23     -0.41  -0.170    -0.170 -0.29    0.032    NA
## att_SA_avg          -2.38     -2.93  -0.594    -0.594 -0.75    0.132    NA
##                  med.r
## att_batt_avg     -0.068
## att_agg_batt_avg -0.170
## att_SA_avg       -0.594
##
##  Item statistics
##                     n raw.r std.r r.cor r.drop  mean    sd
## att_batt_avg     2032 0.706  0.20   NaN  -0.62 0.232 0.195
## att_agg_batt_avg 2032 0.045  0.29   NaN  -0.62 0.099 0.127
## att_SA_avg       2032 0.153  0.66   NaN  -0.27 0.013 0.065
##
## Non missing response frequency for each item
##                     0  0.2 0.333333333333333  0.4 0.6 0.666666666666667 1 miss
## att_batt_avg     0.37 0.00              0.57 0.00   0              0.06 0    0
## att_agg_batt_avg 0.58 0.35              0.00 0.07   0              0.00 0    0
## att_SA_avg       0.96 0.00              0.04 0.00   0              0.00 0    0
```

```r
alpha_attempts_rev <- alpha_attempts %>%
  mutate(att_batt_avg = 1 - att_batt_avg)

alpha(alpha_attempts_rev)
```

```
##
## Reliability analysis
## Call: alpha(x = alpha_attempts_rev)
##
##   raw_alpha std.alpha G6(smc) average_r  S/N   ase mean  sd median_r
##       0.54      0.48    0.51      0.23 0.91 0.011 0.29 0.1     0.17
```

```
##
##  lower alpha upper     95% confidence boundaries
## 0.52 0.54 0.56
##
##  Reliability if an item is dropped:
##                  raw_alpha std.alpha G6(smc) average_r   S/N alpha se var.r
## att_batt_avg         -0.12     -0.15  -0.068    -0.068 -0.13    0.040    NA
## att_agg_batt_avg      0.19      0.29   0.170     0.170  0.41    0.022    NA
## att_SA_avg            0.70      0.75   0.594     0.594  2.93    0.012    NA
##                   med.r
## att_batt_avg     -0.068
## att_agg_batt_avg  0.170
## att_SA_avg        0.594
##
##  Item statistics
##                     n raw.r std.r r.cor r.drop  mean    sd
## att_batt_avg     2032  0.93  0.84  0.77  0.623 0.768 0.195
## att_agg_batt_avg 2032  0.79  0.73  0.61  0.516 0.099 0.127
## att_SA_avg       2032  0.30  0.53  0.12  0.084 0.013 0.065
##
## Non missing response frequency for each item
##                     0  0.2 0.333333333333333  0.4 0.6 0.666666666666667    1
## att_batt_avg     0.00 0.00              0.06 0.00   0              0.57 0.37
## att_agg_batt_avg 0.58 0.35              0.00 0.07   0              0.00 0.00
## att_SA_avg       0.96 0.00              0.04 0.00   0              0.00 0.00
##                  miss
## att_batt_avg        0
## att_agg_batt_avg    0
## att_SA_avg          0
```

```
#std.alpha = .48
#best item = att batt (.93), worst item = .30 att SA
#reliability if att_SA_avg is dropped = .75

alpha_actual <- efa_data_2 %>%
  select(!IDPER)

alpha(alpha_actual)
```

```
## Some items ( SA_avg ) were negatively correlated with the total scale and
## probably should be reversed.
## To do this, run the function again with the 'check.keys=TRUE' option
```

```
##
## Reliability analysis
## Call: alpha(x = alpha_actual)
##
##   raw_alpha std.alpha G6(smc) average_r   S/N   ase mean    sd median_r
##      -0.49     -0.89   -0.45     -0.19 -0.47 0.059 0.21 0.097    -0.18
##
##  lower alpha upper     95% confidence boundaries
## -0.6 -0.49 -0.37
##
```

```
##  Reliability if an item is dropped:
##             raw_alpha std.alpha G6(smc) average_r   S/N alpha se var.r med.r
## bat_avg         -0.44     -0.44   -0.18     -0.18 -0.31    0.100    NA -0.18
## agg_bat_avg     -0.38     -0.71   -0.26     -0.26 -0.42    0.057    NA -0.26
## SA_avg          -0.16     -0.26   -0.12     -0.12 -0.21    0.049    NA -0.12
##
##  Item statistics
##               n raw.r std.r r.cor r.drop  mean   sd
## bat_avg     835 0.901  0.45   NaN  -0.30 0.519 0.3
## agg_bat_avg 835 0.172  0.51   NaN  -0.18 0.085 0.1
## SA_avg      835 0.018  0.41   NaN  -0.32 0.031 0.1
##
## Non missing response frequency for each item
##                0 0.166666666666667 0.333333333333333  0.5 0.666666666666667   1
## bat_avg     0.16              0.00              0.00 0.63              0.00 0.2
## agg_bat_avg 0.56              0.38              0.06 0.00              0.00 0.0
## SA_avg      0.91              0.00              0.08 0.00              0.01 0.0
##             miss
## bat_avg        0
## agg_bat_avg    0
## SA_avg         0
```

```
alpha_actual_rev <- alpha_actual %>%
  mutate(SA_avg = 1 - SA_avg)

alpha(alpha_actual_rev)
```

```
##
## Reliability analysis
## Call: alpha(x = alpha_actual_rev)
##
##   raw_alpha std.alpha G6(smc) average_r  S/N   ase mean   sd median_r
##       0.15      0.27    0.25      0.11 0.37 0.037 0.52 0.12     0.18
##
##  lower alpha upper     95% confidence boundaries
## 0.08 0.15 0.23
##
##  Reliability if an item is dropped:
##             raw_alpha std.alpha G6(smc) average_r   S/N alpha se var.r med.r
## bat_avg          0.31      0.31    0.18      0.18  0.44    0.048    NA  0.18
## agg_bat_avg      0.28      0.42    0.26      0.26  0.71    0.030    NA  0.26
## SA_avg          -0.16     -0.26   -0.12     -0.12 -0.21    0.049    NA -0.12
##
##  Item statistics
##               n raw.r std.r r.cor r.drop  mean  sd
## bat_avg     835  0.89  0.60  0.25  0.094 0.519 0.3
## agg_bat_avg 835  0.25  0.56  0.13 -0.048 0.085 0.1
## SA_avg      835  0.57  0.76  0.58  0.318 0.969 0.1
##
## Non missing response frequency for each item
##                0 0.166666666666667 0.333333333333333  0.5 0.666666666666667
## bat_avg     0.16              0.00              0.00 0.63              0.00
## agg_bat_avg 0.56              0.38              0.06 0.00              0.00
## SA_avg      0.00              0.00              0.01 0.00              0.08
```

```
##                1 miss
## bat_avg     0.20    0
## agg_bat_avg 0.00    0
## SA_avg      0.91    0
```

```
#std.alpha = .27
#best item = batt (.89), worst item = .25 agg_batt
#reliability if agg_bat_avg is dropped = .42
```

Conclusions and Implications:

Demographics and Violent Personal Crime:

Total number of incidents were almost the same for both Male and Female. However, there were a greater number of white respondents reporting being victimised than other races - likely due to overrepresentation in sample. Women were found to be reliably more likely to be sexually assaulted than men, however, the gender difference was not reliable for any other variable of violent crime.

Attempt vs. Actual instance of personal violent crime:

We found a high positive correlation between battery, aggravated battery and attempted sexual assault, but actual sexual assault was negatively correlated. Dimensions in our study 2 factor model chosen, EFA was inconclusive likely due to data selection and cleaning procedure. However, in the regression analysis we saw a great overlap between aggravated battery and battery.

Battery and Aggravated battery are correlated since they are likely to occur together, however, sexual assault is not reported along with battery. That is, someone that was physically assaulted has more charges to file for the same assault (cumulative consequences), but sexual assault is coded separately and receives only one charge. The above conclusions have a larger systematic implication for why sexual assault cases are underreported and harder to prove and win in court, as compared to other personal violent crimes, since they are not treated in the same was as other violent crimes.

Future Directions:

Given more time and resources, we want to look at other demographic variables such as race, socio-economic status, educational attainment and learn how they interact with gender and correlate with instances of personal violent crime. Furthermore, we would like to learn how to conduct these evaluations/EFA after coding missing data NAs so that we don't miss out on a lot of good conclusions. Future studies should also look at non-violent personal crimes and property crimes and analyze their relationship with the demographic variables listed above.

Ethical Reflection

Questions:

1. Are there potential biases in data collection and analysis methodology?
2. Are the conclusions drawn from the data accurate and representative of our demographic? What're some likely misrepresentations of the results?
3. Are we generalizing our conclusions on victimization rates and their relationships with demographic variables based on a snapshot of data from a single year?

4. Are there other variables that could better explain our results that we have missed when deciding the scope of our research?
5. Have certain demographics (race, gender identity) been excluded or combined with others for ease of data analysis, therefore ignoring their unique experiences? Has categorization of demographic variables been harmful to a greater understanding of the relationship between victimization rates and demographics as they exist in the real world?

These questions have been combined and answered below in chronological order of our data analysis procedure.

Research Scope and Bias:

As mentioned at the beginning of this report, this study was limited in its scope and resources, and is biased by the rudimentary knowledge of the researchers on criminal justice and data analysis procedures.

Data Extraction

For ease of data extraction, this study only looked at 2023 data, instead of the available longitudinal concatenated file that contained incident reports over thirty years - therefore this study is only looking at a snapshot of data instead of impact of policy, global events such as the pandemic and recession on victimization rates in the USA.

Moreover, within the 2023 NCVS file, we only extracted incident and person records, rather than analyzing the impact of household and geographical address - which are suggestive of socio-economic status - on the likelihood of victimisation. We also did not consider city vs. urban, income and other variables that interact with demographics of interest to our study to influence victimization rates.

From these two data sets, we manually reviewed the codebook and picked variables we thought would be relevant from the ICPSR data sets - a process that could have led to human error and missed variables that could be relevant to our study.

Data Cleaning/Analysis

Another instance of missed observations include the cleaning procedure. residual and missing data were both coded as NA, even though they have different statistical relevance to research.

Furthermore, since our factor and regression analyses were not robust to NAs, we made the decision to drop all NA values, losing almost 200000 observations. Given more time, we would have conducted demographic analysis of missing data, which could have indicated whether certain demographics were being systematically excluded from our study due to higher rates of missing or NA data, which could be suggestive of bias or discrimination in data collection procedures.

Although we wanted to present gender as more than binary, the available data encoded gender as sex assigned at birth and we were only able to draw comparisons between individuals coded as males and females.

Although we were interested in race as well as gender, we were unable to conduct further analysis of race and its relationship with violent crime. This is because of overrepresentation of white people as compared to other races with lower number of respondents, which would have invalidated our conclusions. Moreover, if the respondent was coded as identifying into multiple races, they were all assigned as “mixed”, and we were uncomfortable drawing unethical inferences about race from this skewed data.

Lastly, our data cleaning procedures led us to inconclusive EFAs and unreliable regression analysis, which affected the internal and external validity of our study.

Data Science Core Competencies Reflection

Leela Addepalli

Collect and analyze data in a reproducible and ethically responsible manner Throughout this final project, as well as through the radiolab podcast discussions over the semester, I believe I have gained a deeper understanding of ethical decision-making in data analysis procedures. For this project, by critically evaluating the pros and cons of each decision, I have been able to justify my thought process and critically evaluate the best possible statistical approach. Learning when to prioritize parsimony over representation and how to buffer missing data by removing observations rather than rows has helped me ensure that minority voices and experiences aren't ignored in my research. Furthermore, identifying my own personal limitations as a research scientist helps me identify areas for future improvement.

Obtain data through searching, scraping, mining or experimental methods We obtained our dataset from the internet, and had to filter out datasets to only include csv files since we had better experience reading those files in. That is how we came upon the NCVS files, and being interested in learning more about the dataset, we chose to analyze it. This part of the process was not very challenging and I believe myself to be fully competent in reading in csv files, however, I do need to build competencies in translating other file formats for R.

Parse, transform and generate wide-ranging data sets for analysis The hardest part about this final project process was cleaning up our data, and not knowing if we were making the right decisions when parsing and organizing our variables until we actually conducted the analysis. This meant that our final EFA and regression analysis were inconclusive, which was initially disappointing. However, I later realized that inconclusive results still have meaning and realizing what those implications were, was very exciting and made me proud of our work. We made full use of joining, filtering, and other functions to transform our dataset when necessary, and I believe myself competent in this area.

Statistically analyze data to summarize, draw inferences and make predictions Another challenging aspect of this process was deciding when to sacrifice certain research directions due to lack of availability of data. An example of this is when we decided to change our research question to look at gender ONLY rather than race and gender, since we did not have equal representation of race. We could have moved forward and conducted data analysis on skewed numbers, knowing that our conclusions would be biased towards certain racial demographics, however, we chose to concentrate our efforts on the demographics that we were more certain were equal. This was also explored in our ethical reflection. I believe we have done a good job summarizing and drawing inferences from the final dataset we were left with.

Identify patterns and relationships in datasets using visualization and algorithms As mentioned above, it was very interesting to realize that even NA and inconclusive numbers have meaning. The one thing we didn't end up doing that I wished I had the time for, is a demographic analysis of missing and NA data. The initial look at the race variable gave me a good indication that the dataset is biased towards some races, and that there is a real possibility that excluding NA and missing data could mean systematically excluding certain demographics from this very important research about the rate of victimization per population. However, I will not forget to prioritise this in future research endeavours. Other than that, I think we did a good job identifying patterns, recognizing our flaws and limitations and rethinking our plans based on what the data showed us.

Communicate data methods and conclusions to diverse audiences I think we did a great job over this semester and throughout this final report and drawing layperson graphs for every visualization. Although our layperson graphs show an average of an average which has limited statistical value, I think they still communicate the essence of our data, which has led us to the conclusions stated above about sexual assault being categorized separate from other violent crimes. Furthermore, I think we did a good job organizing our report logically and sensibly. Hopefully, even though this report is for an expert audience, it should be mostly readable by even a laymen audience.

Overall, I believe this project, as well as this class has helped me get closer to achieving full competency in the six identified core values of the data science program at Washington and Lee University. It has especially left me with many lessons and reminders as I move forward in my research career.